Process Synchronization
— Chapter 6

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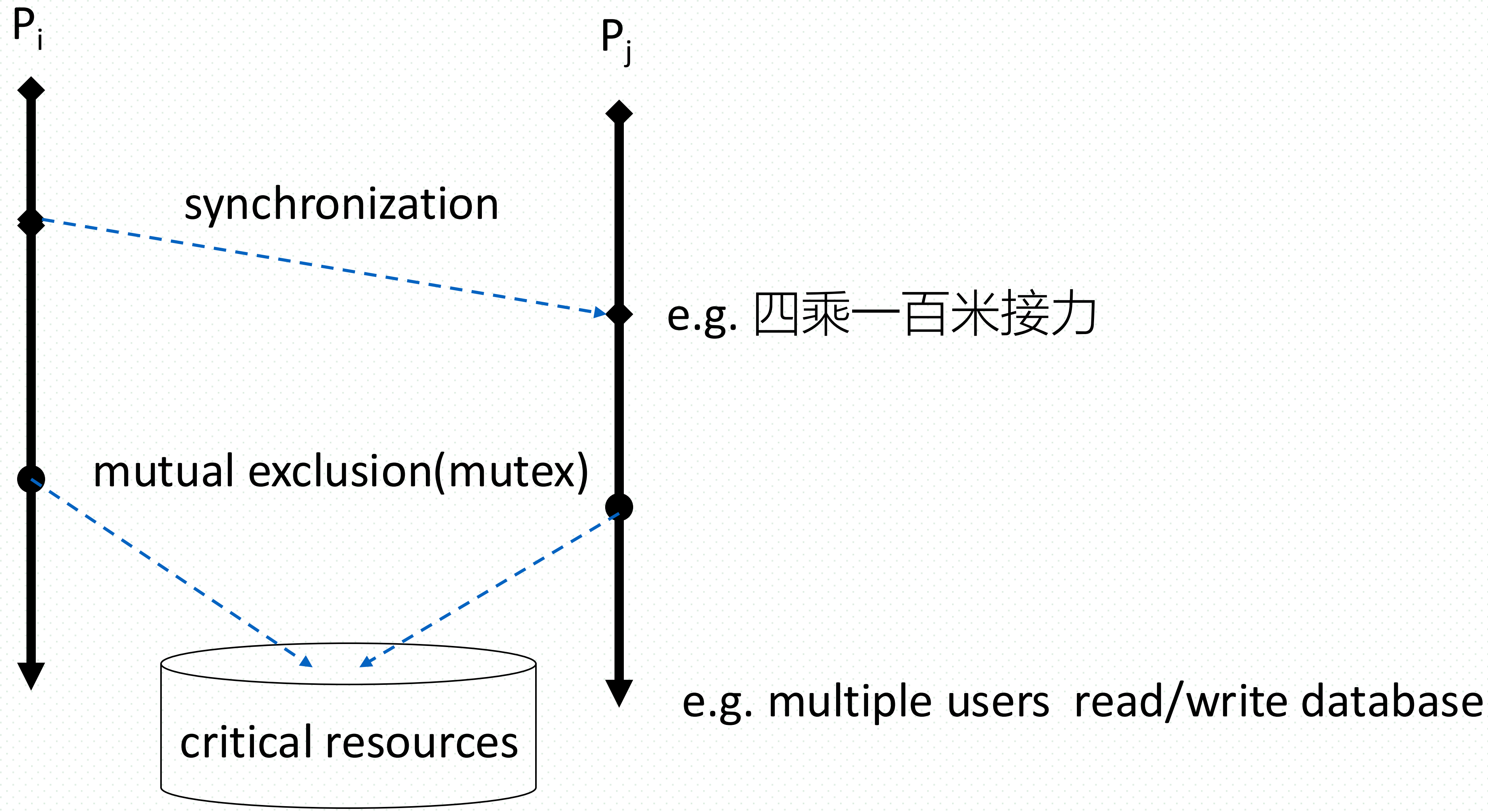
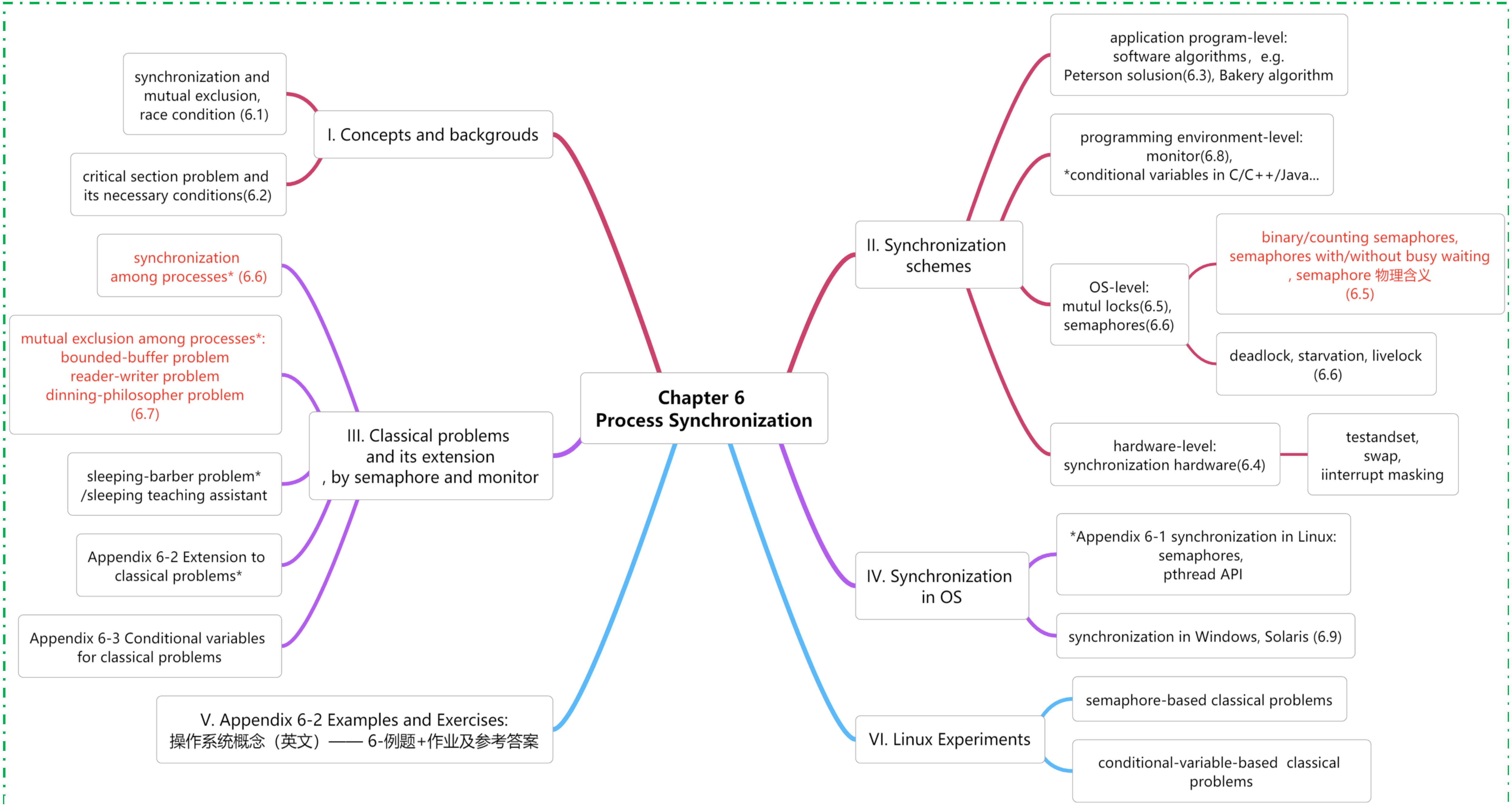
Synchronization vs Mutual Exclusion

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Terminology

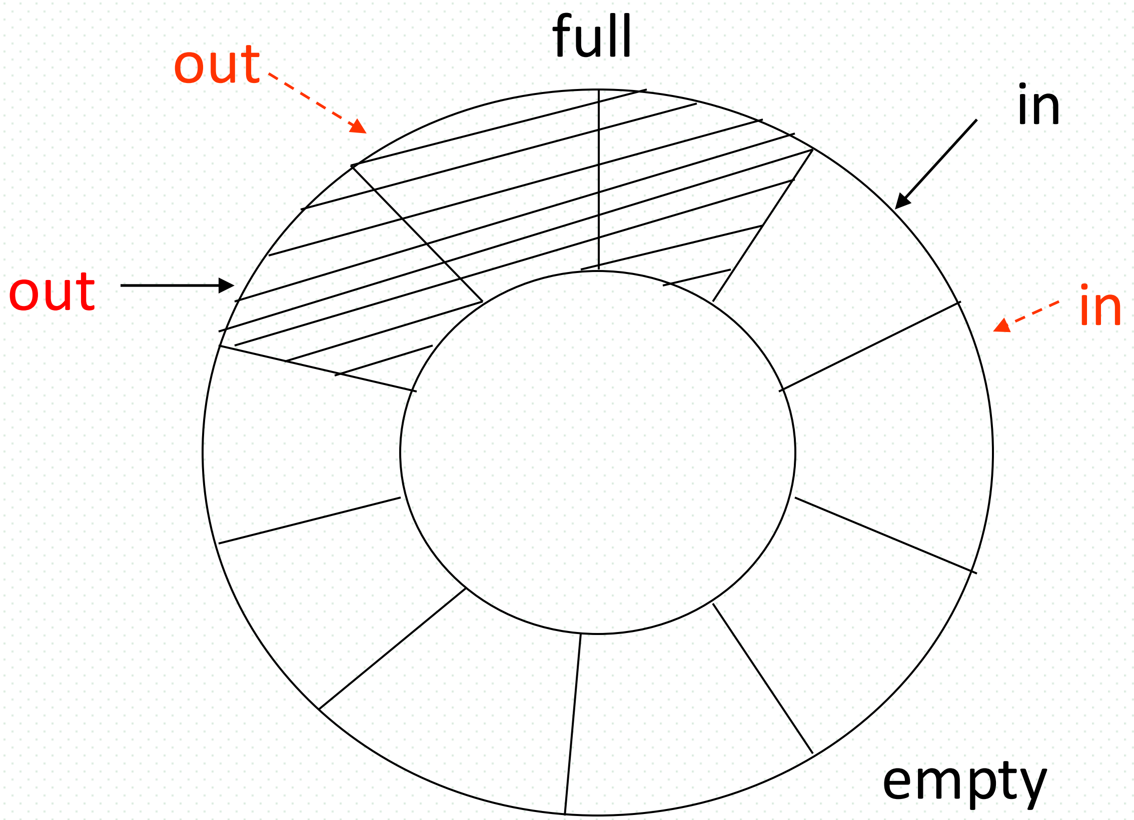
synchronization, mutual exclusion	同步， 互斥
atomic operation	原子操作
interleaving	交织，交叠
race condition	竞争/争用条件
critical section	临界区/段
semaphores, counting semaphore, lock	信号量， 计数(多元)信号量， 锁
deadlock, starvation	死锁, 饿死
busy waiting	忙等待
priority Inversion	优先级反转
monitor	管程
conditional variables	条件变量

6.1 Background

- Cooperating processes vs independent processes in computers (§3.4, P96)
  - process cooperating in four conditions
- **Shared resources**
  - the resources (e.g., data, CPU, I/O ports, memory) that can be accessed by several cooperating processes concurrently
  - the shared resources cannot be used by several processes simultaneously (or in parallel), only be used mutual exclusively(互斥).
- **Shared data** as shared resources
  - concurrent access to shared data in DBS may result in data inconsistency
- Maintaining data consistency requires mechanisms to ensure the orderly execution of cooperating processes
  - process synchronization is needed

Example: Bounded-Buffer Problem

- Also known as Producer-consumer problem
- Synchronization among
  - producers
  - consumers
  - producers and consumers



Bounded-Buffer Problem

- Shared-memory solution to bounded-buffer problem (Chapter 3)
  - allows at most *n* (e.g. 10) items in buffer at the same time
  - adding a variable *counter*, initialized to 0 and incremented each time a new item is added to the buffer

- Codes for bounded-buffer problem

```
#define BUFFER_SIZE 10
        typedef struct {
            . . .
        } item;
        item buffer[BUFFER_SIZE];
        int in = 0; int out = 0; int counter = 0;
```

Bounded-Buffer Problem

- **Producer process**

```
while (1) {
// producing an item in nextProduced */
while (counter == BUFFER_SIZE) ;
/* do nothing */
buffer[in] = nextProduced;
in = (in + 1) % BUFFER_SIZE;
counter++;
}
```

- Consumer process

```
while (1) {
while (counter == 0) ;
/* do nothing */
nextConsumed = buffer[out];
out = (out + 1) % BUFFER_SIZE;
counter--;
/* consume the item in nextConsumed
*/
}
```

Bounded-Buffer Problem

- counter is shared data, the statements

  counter++;

  counter--;

  must be performed *atomically*
- Atomic operation means an operation that completes in its entirely without interruption. Otherwise, data inconsistency will occur.
- An Example: executing of counter++ and counter -- atomically
  - if the producer and the consumer access counter sequentially, data consistency can be maintained
  - e.g. counter=5; counter ++; counter --; the result remains counter=5
  - the statement “count++” and “count--”may be implemented in machine language as

```
register1 = counter
register1 = register1 + 1
counter   = register1
```

```
register2 = counter
register2 = register2 - 1
counter   = register2
```

Bounded-Buffer Problem

- Counter-example: executing of counter++ and counter-- in interleaving manner
  - both the producer and consumer attempt to update the buffer concurrently, the assembly language statements may get interleaved(交织)
  - interleaving depends upon how the producer and consumer processes are scheduled, and may result in *inconsistent data*
  - assume **counter** is initially 5. One interleaving of statements is

```
producer:   register1 = counter           (register1 = 5)
producer:   register1 = register1 + 1     (register1 = 6)
consumer:   register2 = counter           (register2 = 5)
consumer:   register2 = register2 – 1     (register2 = 4)
producer:   counter   = register1         (counter = 6)
consumer:   counter   = register2         (counter = 4) .
```

  - Problem
    - the value of count may be either 4 or 6, where the correct result should be 5

Why synchronization needed: Race Condition

- Race condition(竞争条件/竞态条件)
  - the situation where several processes access and manipulate shared data concurrently
  - the final value of the shared data depends upon which process finishes last
    - 结果不唯一、不确定
- To prevent race conditions, concurrent processes must be synchronized, i.e. access shared resources in mutual exclusory methods

6.2 The Critical-Section Problem

Some concepts related to Critical-Section Problem

- *n* processes, all competing to use some shared data/resources
- **Critical section (临界区/段)**
  - **e**ach process has a code segment, called critical section, in which the shared data is accessed.
- **Critical resources**
  - resources accessed by critical section

- **Requirement** – ensure that
  - when one process is executing in its critical section to access critical resources, no other process is allowed to enter its critical section to access the same critical resources

General Structure of Process

- General Structure of Process

```
do {
      …….
      entry    section(R)   // apply
      critical section(R)   // use
      exit     section(R)   // release
      reminder section
} while (1);
```

Solution to Critical-Section Problem

- Necessary conditions for correct solutions to critical-section problems
  - mutual exclusion
  - Progress(进展性)
  - bounded waiting(有限等待)

- **Mutual exclusion**
  - if process P_i is executing in its critical section *specific to the resource R*, then no other processes can be executing in their critical sections *specific to R*
    - /*互斥地执行临界区代码
  - what is mutual exclusion (互斥)
    - a programming technique ensuring that only one program or routine *at a time* can access some resources, such as a memory, an I/O port, or a file

Mutual Exclusion

- Suppose Process P_1 and P_2 concurrently access resource R_1 (e.g. main memory) and R_2 (e.g. I/O devices), so each process has two critical sections responding to R_1 and R_2 respectively
- P_1 and P_2 may be simultaneously stay in its critical sections, however access different resources, i.e. R_1 and R_2

P_1	P_2
entry section(R_1)	entry section(R_2)
critical section(R_1)	critical section(R_2)
exit(R_1)	exit(R_2)
entry section(R_2)	entry section(R_1)
critical section(R_2)	critical section(R_1)
exit(R_2)	exit(R_1)

Progress

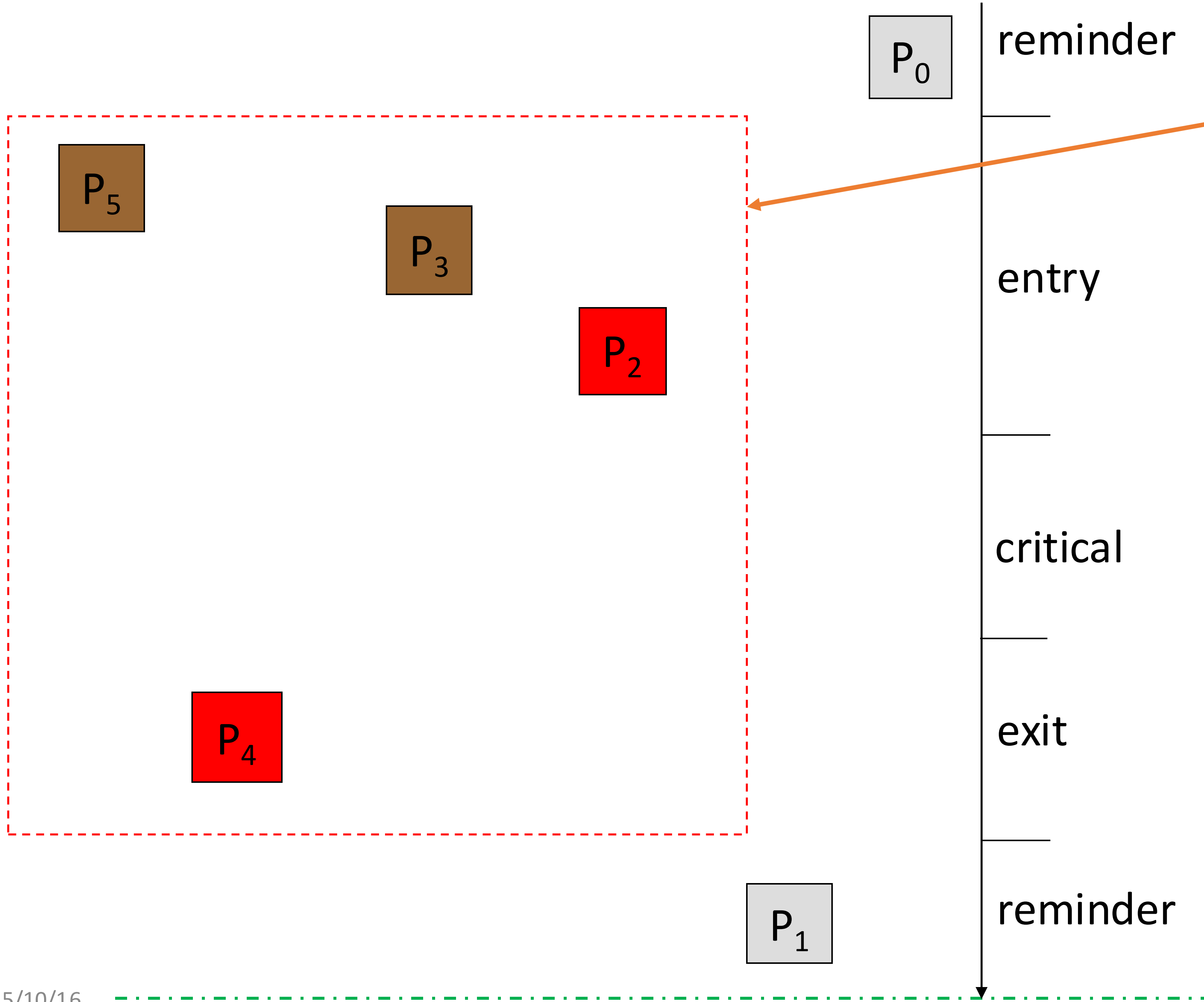
- **Progress**
  - if no process is executing in its critical section (i.e. 共享资源空闲) and there exist some processes that wish to enter their critical section, then
    - (1) *only those processes that are not executing in their reminder sections can participate in the decision* on which will enter its critical section next, and

      // 挑选进入临界区的进程时，不考虑那些正在reminder 区中 执行的进程(已经使用完临界资源)；

      或：只考虑那些正处于entry(申请进入)、critical sections和exit sections 的进程

      临界区外的进程不应阻塞其它进程
    - (2) this selection *cannot be postponed indefinitely*

      // 挑选过程不应被无限期阻止



participate in the decision on which will enter its critical section next

左边方框中的四个进程P_2,P_3,P_4,P_5中，P_2,P_3,P_5处于进入段，提出资源访问请求。P_4处于退出段，正在释放资源。其它两个进程P_0,P_1处于剩余段，与资源访问无关；没有进程处于临界段，资源空闲可用。

同步互斥机制判断挑选进程进入临界段访问互资源时，只考虑进程P_2,P_3,P_4,P_5。其中，进程P_2,P_3,P_5有访问资源需求。进程P_4正在释放资源，只有释放之后，其它三个进程才有可能访问资源

Bounded Waiting

- Bound waiting
  - a bound or limit must exist on the number of times that other processes are allowed to enter their critical sections after a process has made a request to enter its critical section and before that request is granted
    - 当一个进程申请进入临界区并还未得到许可时，应当限制其它进程进入临界区的次数。否则，会导致该进程在临界区外无休止等待——进程在临界段内只应停留有限时间
  - to avoid deadlock or starvation

- Counter-example
  - priority-based preemptive CPU scheduling, the process with lower priority may be starved

进程互斥使用临界资源的原则

- 在共享同一个临界资源的所有进程中，每次只允许一个进程处于它的临界段中
- 若有多个进程同时要求进入它的临界段，应在有限时间内让其中之一进入，而不应相互阻塞使得各进程都无法进入临界段
- 进程在临界段内只应停留有限时间
- 不要使需要进入临界段的进程无限制地等待在临界段之外
- 处于剩余段的进程不应阻止其他进程进入临界段

6.3 Peterson Solution

- In user mode，a software-based solution to process synchronization

- G.L.Peterson (1981) algorithm
  - for synchronizing two processes

- Bakery Algorithm 【略】
  - critical section solution for n>=2 processes

Peterson Solution

- the structure of process P_i

```
enter_section ( i );    //判断是否可安全进入，如不能，等待
critical_section(i);
exit_section ( i );     //退出时，应允许其它进程进入临界段
reminder section (i);
```

- For processes P_i and P_j , shared variables:
  - **boolean flag [2]**;

    /* **flag [i] = true** $\Rightarrow P_i$ ready/want/apply to enter its critical section */

    initially **flag [i] = flag [j] = false.**

  - **int turn**; /*轮到谁？*/

    turn == i $\Rightarrow P_i$ are allowed to enter its critical section

    nitially **turn = 0**

- Process P_i

```
        do {
                 flag [i]= true;       /*表明自身请求进入临界区*/
                    turn = j;          /* 假设先轮到对方进入临界区 */
enter_section    while (flag [j] and turn == j)
                   {  do-nothing  };
                                        /*对方正在临界区中, Pi忙等待*/
                  critical section
exit_section      flag [i] = false      /*表明自身不再需要进入临界区*/
                  remainder section
        } while (1);
```

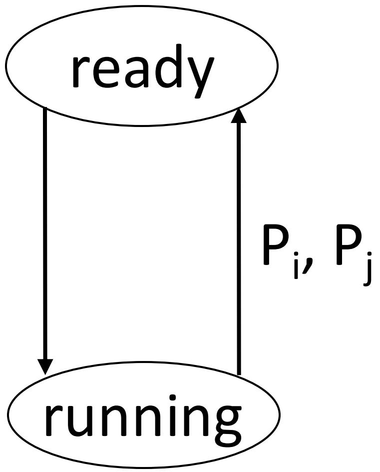
- 当P_i和P_j同时申请访问临界资源时，微观上先申请者（先置位turn）先进入临界区， 假设： 单CPU系统

P_i	P_j	timeline
flag [i]= true		↓
	flag [j]= true	
turn = j;		
	turn = i;	
false: flag [j] and turn == j		
	true: flag [i] and turn == i	
critical section i	*busy waiting*	

Peterson Solution (cont.)

- Features
  - **this algorithm is correct**
    - meets requirements of mutual exclusion, progress, bounded-waiting;
    - a solution to critical-section problem for two processes
  - demerit
    - **busy-waiting**

  P_i alternatively stays in running and ready states processes pass two states:



It is assumed that:

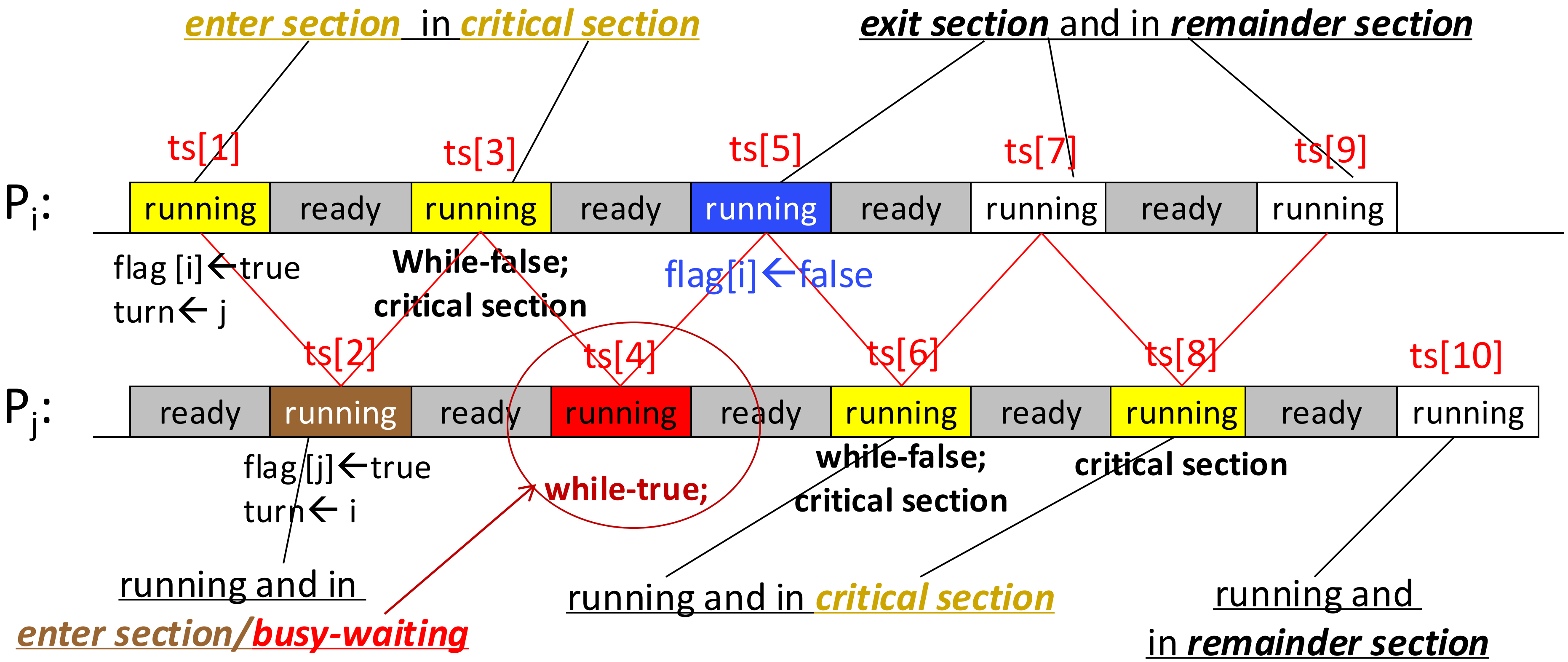
1. P_i is in its critical section, and P_j is in the busy- waiting state
2. time-sharing scheduling:
   time slots ts[i](e.g. 50ms) are assigned to P_i and P_j alternatively

```
while (flag [j] and turn == j)



while (flag [i] and turn == i)
```

busy-waiting



Bakery Algorithm【略】

- An software solution for synchronize among ***n* processes**
- Principles
  - before entering its critical section, process receives a number (denoted for ticket #)
  - the holder of the smallest number enters the critical section
  - if processes P_i and P_j receive the same number, if $i < j$, then P_i is served first; else P_j is served first
  - the numbering scheme always generates numbers in increasing order of enumeration; i.e., 1,2,3,3,3,3,4,5...

Bakery Algorithm (cont.)

- Notes : for lexicographical order (ticket #, process id#)
  - $(a,b) < (c,d)$ if $a < c$ or if $a = c$ and $b < d$
  - **max** $(a_0,\ldots, a_{n-1})$ is a number k, such that $k \geq a_i$ for $i = 0,\ldots, n-1$

- Shared data
  - **boolean choosing[n];**

    /* choosing[i]=true: P_i begins/wants to apply for ticket*/
  - **int number[n];** /* number[i]: ticket# for P_i ,

    number[i]>0 表示P_i需要进入临界区: */
  - data structures are initialized to **false** and **0** respectively

Process P_i

time slots (e.g. 50ms) are assigned to processes alternatively

```
do {
   choosing[i] = true;          /*begin to apply for ticket*/
   number[i]
     = max(number[0], number[1], …, number[n – 1]) + 1;
                    /* a ticket whose number is maximum  is given ,
                     first come first given, in non-decreasing order */
   choosing[i] = false;         /* end application */

for (j = 0; j < n; j++)         /*考虑所有进程*/
{
     while (choosing[j]) ;    /*如有其它进程正在申请ticket, 等待Pj申请完*/

     while ((number[j] != 0) &&  {(number[j], j) < (number[i,], i)}) ;
                    /*如有票号小于自身的其它进程Pj 正在或等待运行,
                     等待Pj访问完*/
}
```

P_i和P_j两个进程，采用时间片轮转，申请票号；P_i在第一个时间片内读取其他进程的最大number=5，在还没有完成通过max操作将number[i]设置为6的时候，CPU切换到P_j，P_j读取到的其他进程最大值仍然是number=5，通过max获取的票号也将是6

```
    critical section        /*进入临界区, 访问临界资源*/
    number[i] = 0;          /*丢弃使用过的票，退出临界区*/
    remainder section
} while (1);
```

- Features
  - Bakery algorithm is correct
  - demerit: busy-waiting
    - **while**

6.4 Synchronization Hardware 【略】（例题）

45.（7 分）下表给出了整型信号量 S 的 wait( )和 signal( )操作的功能描述，以及采用开/关中断指令实现信号量操作互斥的两种方法。

功能描述	方法 1	方法 2

Semaphore S;
wait(S) {
while(S <= 0);
S = S-1;
}
 	
带有忙等待信号量
	Semaphore S;
	wait(S) {
	关中断;
	while(S <= 0);
	S = S-1;
	开中断;
	}
		Semaphore S;
		wait(S) {
		关中断;
		while(S <= 0) {
		开中断;
		关中断;
		}
		S = S-1;
		开中断;
		}

signal(S) {
S = S+1;
}
	signal(S) {
	关中断;
	S = S+1;
	开中断;
	}
		signal(S) {
		关中断;
		S = S+1;
		开中断;
		}

此刻，其它进程插入，修改S

请回答下列问题。

（1）为什么在 wait( )和 signal( )操作中对信号量 S 的访问必须互斥执行？

（2）分别说明方法 1 和方法 2 是否正确。若不正确，请说明理由。

（3）用户程序能否使用开/关中断指令实现临界区互斥？为什么？

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45.【7 分】现要求学生使用 swap 指令和布尔型变量 lock，实现临界区互斥。lock 为线程间共存的变量。lock 的值为 true 时线程不能进入临界区。为 false 时线程能进入临界区。某同学编写的实现临界区互斥的伪代码如题 45（a）所示

某同学写的伪代码	newswap ( ) 的代码
bool lock=FALSE;//共享变量	void n ewswap(bool*a,bool*b)
//进入区	{
bool key=TRUE	bool temp=*a;
if(key)=TRUE	*a=*b
swap key,lock;//交换 key 和 lock 的值	*b=temp
//临界区	}
lock =TRUE 推出区	

题 45 (a) 图　　题 45 (b) 图

（1）题 45（a）图中伪代码中哪些语句存在错误，进行改正，不增加语句条数。

（2）题 45（b）图中给出了两个变量值的函数 newswap（）的代码是否可以用函数调用语句“newswap(&key,&lock)”代替指令“swapkey,lock”以实现临界区的互斥？为什么？

Synchronization Hardware

- Approaches for the critical-section problem, i.e. synchronization, or mutual exclusion, among concurrent processes, to avoiding race condition
  - In single-processor system
    - interrupt masking/disabling, 中断屏蔽，关中断
  - In multiple-processor system
    - Test-and-Set instruction
    - Compare-and-Swap instruction

Interrupt Masking in Uniprocessor Systems

- In a single-processor environment, the critical-section problem could be solved by
  - preventing interrupts from occurring while a shared variable was being modified

关中断/中断屏蔽
进入临界段
访问共享变量
退出临界段
开中断

- 注：进程P_i关中断后，禁止了用户模式-内核模式切换，其它进程P_j无法抢占临界区内的P_i，从而在P_i访问完共享变量V、退出临界区前，P_j无法访问共享变量V，从而避免race-condition
- In this way, we could be sure that the current sequence of instructions (in critical section) would be allowed to execute in order without preemption.
- No other instructions would be run, so no unexpected modifications could be made to the shared variable. This is often the approach taken by nonpreemptive kernels.

Interrupt Masking in Uniprocessor Systems

- Interrupt masking based hardware synchronization
  - 进入临界区前执行："关中断"指令
  - 离开临界区后执行：执行"开中断"指令
  - "开关中断"指令保证用户态下的进程在临界区执行时不被其它进程中断，从而实现多个进程互斥访问临界区
- 特点
  - 简单，有效
  - 较高的代价，限制CPU并发能力(临界区大小)
  - 不适用于多处理器

Synchronization Hardware in Multiprocessor Systems

- **Disabling interrupts** on a multiprocessor can be time consuming, since the message is passed to all the processors. This message passing *delays* entry into each critical section, and system efficiency decreases【延缓进程进入临界区的速度】
- Also consider the effect on a system's clock if the clock is kept updated by interrupts
- Many modern computer systems therefore provide special hardware instructions that allow us either to test and modify the content of a word or to swap the contents of two words **atomically**—that is, as one uninterruptible unit.
- We can use these special instructions to solve the critical-section problem in a relatively simple manner
- In multiple-processor environments, two schemes of hardware synchronization by machine instructions
  - Test and Modify
  - Compare and Swap

Synchronization Hardware in Multiprocessor Systems

- Principles of TestAndSet Instruction and CompareAndSwap Instructions
  - 每个临界资源设置一个布尔变量lock，初值为false，lock用机器字实现
  - CPU提供专门的硬件指令TestAndSet 或CompareAndSwa，允许对一个字的内容进行检测和修正，或交换两个字的内容
  - 硬件指令可以解决共享变量的完整性和正确性，防止出现race condition
  - 进程在enter_section中利用原子操作TestAndSet 指令或 Swap 指令测试lock, 察看临界资源是否空闲, 从而决定是否进入critical section

- Properties
  - 简单、有效，特别适用于多处理机
  - 缺点：忙等待

Test_and_Set Instruction

- Test and modify the content of a word **atomically**, by ***TestAndSet*** instruction
  - /* 测试布尔变量target的值，并返回该值，再将target设置为true

```
boolean TestAndSet(boolean  *target) {
        boolean rv = *target;
        *target = true;
        return rv;
    }
```

  - Executed atomically
  - Returns the original value of passed parameter
  - Set the new value of passed parameter to “TRUE”.

- If two TestAndSet instructions are executed simultaneously (each on a different CPU), they will be executed sequentially in an arbitrary orders

Test and Set

- To conduct mutual exclusion on a type of resource, declare a Boolean variable ***lock***, specific to the resource and initialized to *false*

- The processes utilize ***TestAndSet*** instructions on this *lock* to mutual exclusively access the resource, as follows
  - shared data

    boolean *lock* = false   /*表示没有加锁，无用户访问

Process P_i,

```
do {
   while (TestAndSet(lock)) ;
     /* false→true, 加锁，阻止其它进程
    critical section

    lock = false;
    /*释放锁，允许其它进程进入
   remainder section
}
```

```
TestAndSet (lock)
  { boolean rv = lock
     lock = true;
     return rv;
   }
```

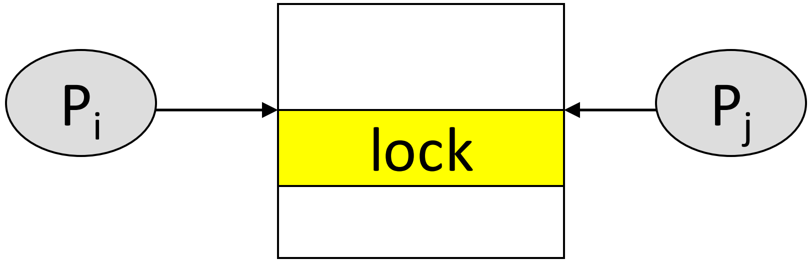
P_j

```
do {
  while (TestAndSet(lock)) ;
   /*busy-waiting

  while (TestAndSet(lock)) ;
     /* false→true, 加锁，阻止其它进程)
critical section

 lock = false; /*释放，允许其它进程进入

 remainder section
}
```



lock

F

T

F

T

F

Compare_and_swap Instruction

- Definition:

```
int compare_and_swap(int *value, int expected, int new_value) {
                           老值         期望值              新值
    int temp = *value;
    if (*value == expected)
       *value = new_value;
 return temp;
}
```

  - Executed atomically
  - Returns the original value of passed parameter “value”
  - Set the variable “value” the value of the passed parameter “new_value” but only if “value” ==“expected”. That is, the swap takes place only under this condition

Solution using Compare_and_swap() Instruction

- A global variable (lock) is declared and is initialized to 0
  - 代表资源空闲
- The first process that invokes compare and swap() will set lock to 1—表示占用资源. It will then enter its critical section, because the original value of lock was equal to the expected value of 0.
- Subsequent calls to compare and swap() will not succeed, because lock now is not equal to the expected value of 0.
- When a process exits its critical section, it sets lock back to 0, which allows another process to enter its critical section.
- The structure of process P_i is shown in Figure 6.6.

Solution using Compare_and_swap() Instruction

- Shared integer “lock” initialized to 0;
- Solution:

```
do {
    while (compare_and_swap(&lock, 0, 1) != 0) ;  /老值lock!=0，说明资源被占*/
     /* do nothing */                           /老值lock=0，设置为1，占用资源，进入临界区
    critical section
    lock = 0;                                  /*释放资源
   remainder section
} while (true);
```

bounded-waiting mutual exclusion with TestAndSet

Process P_i

- Shared data (initialized to **false**):

boolean lock;

boolean waiting[n];

- 1

```
do {
   waiting[i]=true;
 key = true;
   while (waiting[i]&&key )
        key=TestAndSet(lock);
  waiting[i]=false;
        critical section
   j=(i+1) %n;
  while ((j!=i) && !waiting[j])
   j = (j+1) %n
  if  (j==i)
     lock=false
  else
     waiting[j]=false;
   remainder section
}
```

/*进入段，lock, waiting[i]=true时，资源被占，陷入while:等待

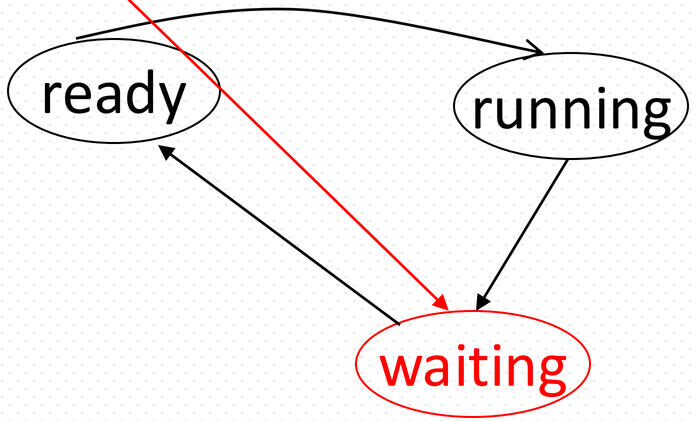
表明自身不再需要等待

考察并释放其它等待进程,使这些进程跳出while循环

6.5 Semaphores

- Semaphore definitions
  - synchronization tools that ***may*** *avoid busy waiting*
  - provided by OS, working in kernel space, used by users through system calls wait()/signal(), etc.
    - 由操作系统在内核空间提供的一种用于多个合作进程间同步与互斥的机制
  - **二元信号量**也被称为锁(lock)，用于进程**互斥，也可用于进程间同步**

- Representation of Semaphore *S*
  - integer variable in kernel space
    - *binary* or *non-binary*
  - two standard operations modify *S*:
    wait() and signal()
    , originally called P() and V() [早期教科书中]



Semaphore Types

- **Binary** semaphore (二元信号量)
  - integer value can range only between 0 and 1, be simpler to implement
    - 用于进程间互斥
  - also known as mutex lock (互斥锁)
- **Counting** semaphore (计数、多元、一般信号量)
  - integer value can range over an unrestricted domain
    - 用于进程间同步、互斥
  - the value corresponds to the number of shared resources usable
- 同步：相互协作多个进程间，由于需要协调它们的工作而相互等待，或需要相互交换信息而产生的制约关系
- 互斥：进程间因相互竞争使用临界资源而产生的制约

Semaphore Features

- Semaphore with busy waiting
  - processes pass only ready and running states
- Semaphore without busy waiting
  - processes pass three states: ready, running, waiting
  - when the resources represented by the semaphores are not available, the processes enter into the waiting state

Semaphores with Busy Waiting

- Operations/system calls on semaphore S
  - **initialization** (S)
    - initialize S to a value
  - **wait** (S) {

    while S ≤ 0 do no-op;

    /* the process that issues

    this operation is waiting*/

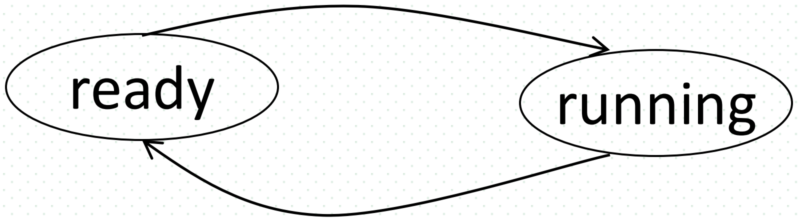
    S--;

    }
  - **signal** (S) {

    S++;

    }

processes pass two states:



time-sharing scheduling, time slots (e.g. 50ms) are assigned to processes alternatively

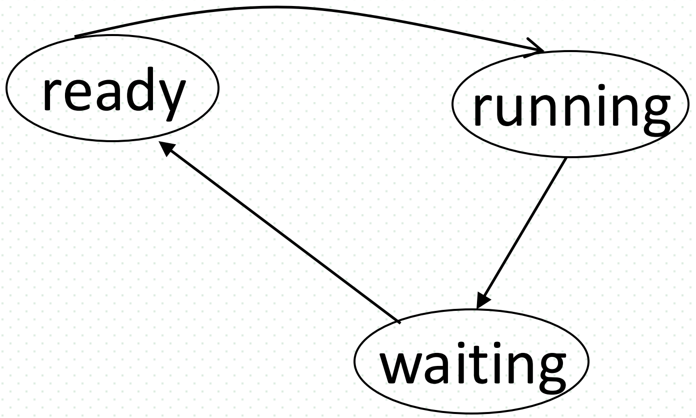
- Wait() may results in process busy waiting
- Example
  - 自旋锁Spinlock in Linux

Semaphores without Busy Waiting

- Define a semaphore S as a C struc

```
typedef struct {
    int       value;
    struct  process *list;
                        /*因S而阻塞/等待的进程队列*/
} semaphore;
```

processes pass three states:
ready, running, waiting



- Assume two simple operations
  - **Block**
    - suspends the process that invokes it, i.e., the process are changed into ***blocked*** **(i.e. *waiting*)** ***state***
  - **wakeup(*P*)**
    - resumes the execution of a blocked process **P,** i.e. from the waiting to ready state

Semaphores Without Busy Waiting

- Operations on <u>*counting* (non-binary)</u> semaphore defined as

  - **wait(S)**{

```
S.value--;  /*申请一个单位的资源/
if (S.value < 0) { /*无可用资源*/
    add this process to S.list;
    block();
}
}
```

  - compared with busy-waiting wait()

  - **signal(S)** {

```
S.value++;        /*释放一个单位的资源*/
if (S.value <= 0)       /*仍有被阻塞进程*/
    { remove a process P from S.list;
      wakeup(P);     /*唤醒被阻塞进程*/
    }
}
```

  - compared with busy-waiting signal()

Semaphores Without Busy Waiting

- Operations on *binary* semaphore defined as
  - **waitB(S)**{

    if S.value=1

    then S.value=0;

    else { add this process to **S.L**

    block();

    }

    }
  - **signalB(S)** {

    if S.L is empty   / there is no process blocked and S.value is 0

    the   S.value=1

    else { remove a process P from S.L;

    wakeup(P) /*唤醒被阻塞进程*/

    }

    }
  - compared with busy-waiting wait()
- A counting semaphore S can be implemented as a binary semaphore and an integer variable

信号量S物理意义

- 针对无忙等待的多元信号量
  - processes pass three states, i.e. ready, running and waiting
- S对应于临界/共享资源
  - S (>0) ——可用共享资源的数目

    S (=< 0) —— 无共享资源可用，|S|表示因请求使用S而被**等待或阻塞**的进程 (注：针对不带有忙等待的信号量)
- wait()
  - 请求分配一个单位的资源S给执行wait操作的进程
- signal()
  - 进程释放一个单位的资源S

- Wait and signal in practical OS
  - in Windows, WaitForSingleObject()

    ReleaseSemaphore()
  - In Linux, down(), up()

说明
Linux提供了可以对信号量进行加、减n(n>1)的wait/signal操作,
允许进程一次申请、释放多个资源实例，简化了基于信号量的编程

Example

- Considering n=10 processes, which mutual exclusively use the resource type *R* of m=6 instances
- A semaphore *S* is designed to synchronize these processes. The maximum and minimal values of *R* are ______respectively
  - A. 10, -4 B. 6, -4

    C. 10, 6 D. 6, -10

value range of S： [m-n, m], e.g. n=10, m=6

Edsger Wybe Dijkstra 与P/V Operations on Semaphores

- Edsger Wybe Dijkstra (1930-2002)
  - 出生于荷兰，美国Texas大学计算机科学和数学教授
  - Dijkstra单源最短路径算法的发明者
  - Turing Prize Winner (1972)
    - 因ALGOL第二代编程语言，获得图灵奖
  - "Go To Statement Considered Harmful"(EWD215): 被广为传颂的经典之作
- 互斥信号量及P/V操作
  - 1962/1963年首次提出，Dijkstra及其团队为Electrologica X8开发一款操作系统——称为THE多道程序系统，提出用P、V操作实现进程的同步与互斥
  - P和V分别取自荷兰语的测试 (**P**roberen) 和增加 (**V**erhogen) 的首字母
    - 也称为wait、signal操作
  - 最初提出的是二元（互斥）信号量，后推广到一般信号量（多值，同步）

Edsger Wybe Dijkstra

- 1965年，提出并解决Dining-Philosophers Problem
  - 多个并发进程竞争使用多个资源
- 死锁处理与Banker’s Algorithm
  - 首次提出死锁概念及术语“deadly embrace”
  - 1965年，为THE系统开发死锁处理策略，提出用于死锁避免的银行家算法
    - 以银行借贷系统的分配策略为基础，判断并保证系统的安全运行

不允许用户应用程序在用户态下直接对信号量操作

- 信号量是OS提供的一种内核空间进程同步互斥机制
- 在用户态下运行的用户应用程序只能通过wait()、signal()等系统调用，**间接地**操纵修改信号量
- 不允许用户应用程序直接在用户态下修改信号量
- 反例

```
defining  semaphore S  integer
          main{
            ...
            S := S – 1
           if  S > 0 then ...
                    else  ...
          }
```

信号量三种用法

- 1. 资源互斥使用
  - 资源只有1个实例，各个进程通过二元互斥信号量mutex互斥地进入临界区，使用资源
    - mutex：代表资源的控制权，初值为1
  - e.g. 生产者-消费者问题中的buffer
- 2. 资源竞争使用
  - 资源有多个实例，允许多个进程竞争使用资源
  - 多元信号量表示：1)资源可用数目，2) 资源使用权
  - e.g. empty, full in the bounded-buffer problem
- 3. 进程间同步——作用类似于条件变量
  - 进程间的执行步骤需要有先后顺序关系
  - 同步二元信号量sync，初值为0

Usage in Mutual Exclusion

- **Mutual exclusion** for *n* processes to concurrently access resource R
  - shared data/resource R by *n* processes

    binary semaphore **mutex**;   /* corresponding to shared data/resource,

    initially mutex = 1

    针对共享资源R，
    设立mutex，代表对资源的
    控制权/使用权

  - Process P_i

```
do {
    wait(mutex) ;
    critical section
    signal(mutex) ;
    remainder section
} while (1);
```

Usage in Synchronization

- **Synchronizing** of *n* processes
  - concurrent process P_1 with statement S_1 and process P_2 with statement S_2
  - requirement: S_2 should be executed only after S_1 has completed
  - P_1 and P_2 share semaphore **synch (表示S_1是否已经执行完)** which is initialized to 0
  - P_1: M_1;

```
         S1;
         signal(synch);
   P2: M2;
         wait(synch);
         S2
```

Example One

- Consider processes P_1, P_2, and P_3 in the following figures.
  - in Fig. (a), only after P_1 ends, can P_2 begin. And only after P_2 ends, can P_3 begin;
  - in Fig. (b), only after both P_1 and P_2 end, can P_3 begin;
  - in Fig. (c), only after P_1 ends, can both P_2 and P_3 begin
- Define semaphores, and synchronize the execution of P_1, P_2, and P_3 by wait() and signal() on these semaphores. Assuming the main bodies of P_1, P_2, and P_3 are as follows

P_1:	P_2:	P_3:
begin	begin	begin
S_1;	S_3;	S_5;
S_2;	S_4;	S_6;
end	end	end

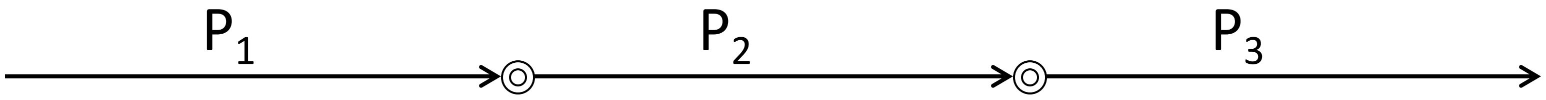


Fig. (a)

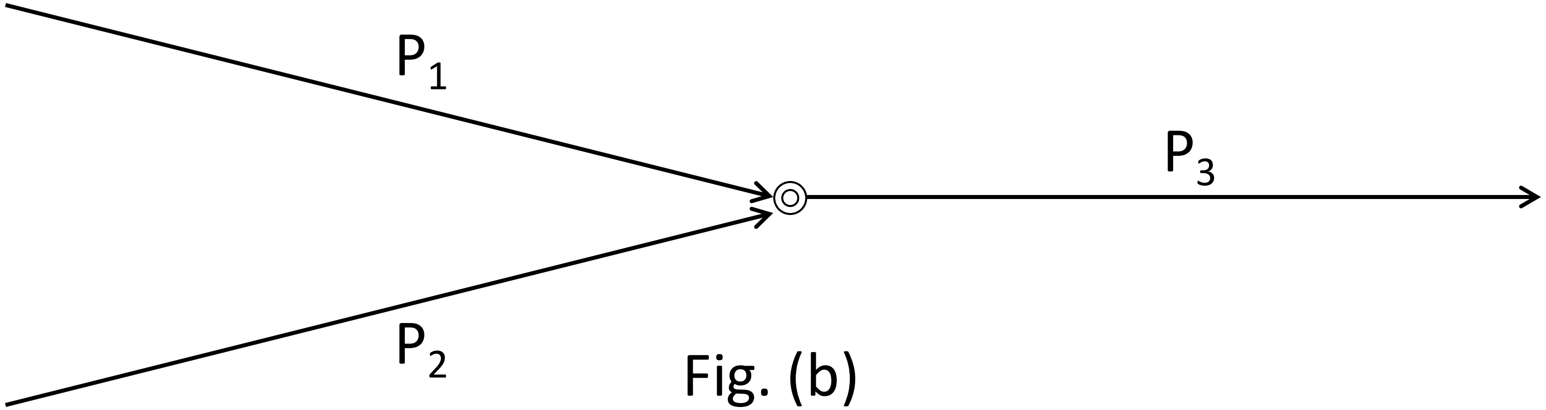


Fig. (b)

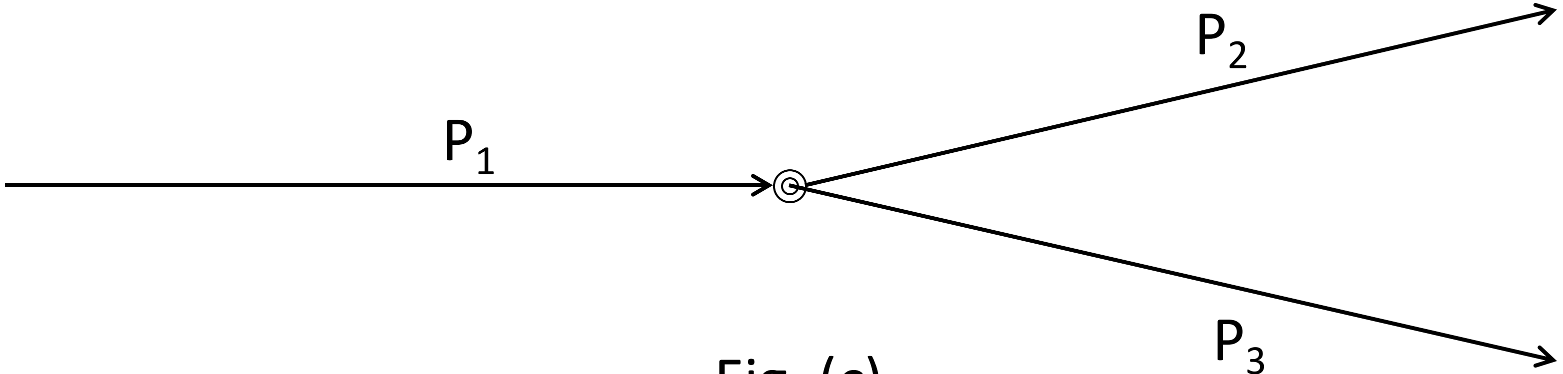


Fig. (c)

Example One

- Initially, defining and initializing **binary** semaphores S1 and S2 as:
  - (binary) semaphores sync1, sync2;
  - sync1:=0, sync2:=0;
- For processes in Fig. (a)

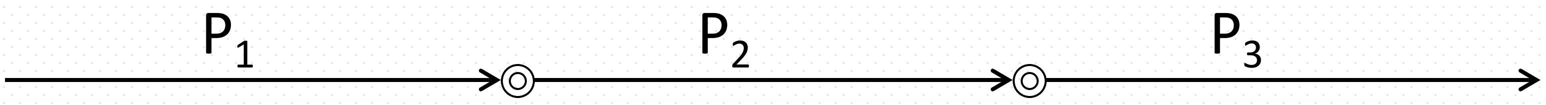


Fig. (a)

P_1:

```
begin
 S1;
 S2;
signal(sync1);
end
```

P_2:

```
begin
 wait(sync1);
 S3;
 S4;
 signal(sync2);
 end;
```

P_3:

```
begin
 wait(sync2);
 S5;
 S6;
 end
```

Example One

- For processes in Fig. (b)

P_1:	P_2:	P_3:
begin	begin	begin
S1;	S3;	wait(sync1);
S2;	S4;;	wait(sync2);
signal(sync1);	signal(sync2);	S5;
end	end	S6:
		end

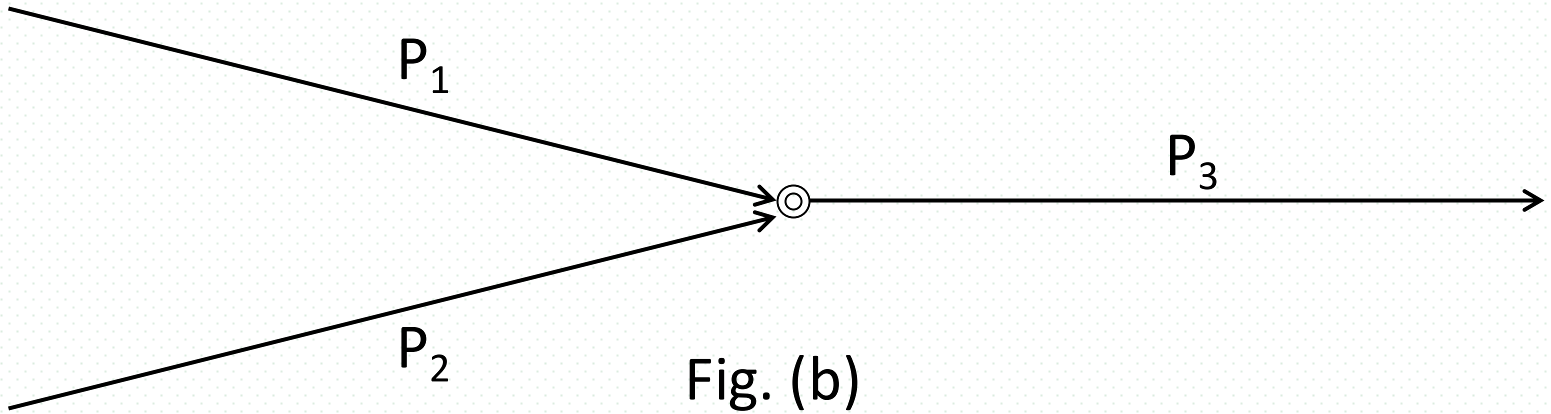


Fig. (b)

Example One

- For processes in Fig. (C)

P_1:	P_2:	P_3:
begin	begin	begin
S1;	wait(sync1);	wait(sync2);
S2;	S3;	S5;
signal(sync1);	S4;	S6:
signal(sync2);	end	end
end		

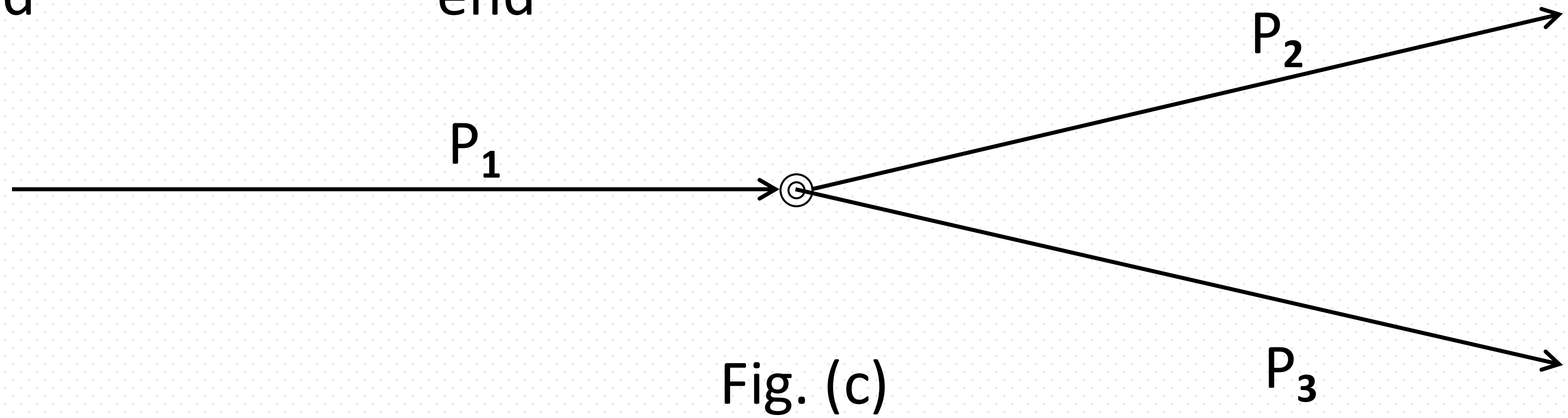


Fig. (c)

Example Two

- 用信号量描述**4** 乘**100** 米接力过程
  - (binary) semaphores S1, S2, S3;

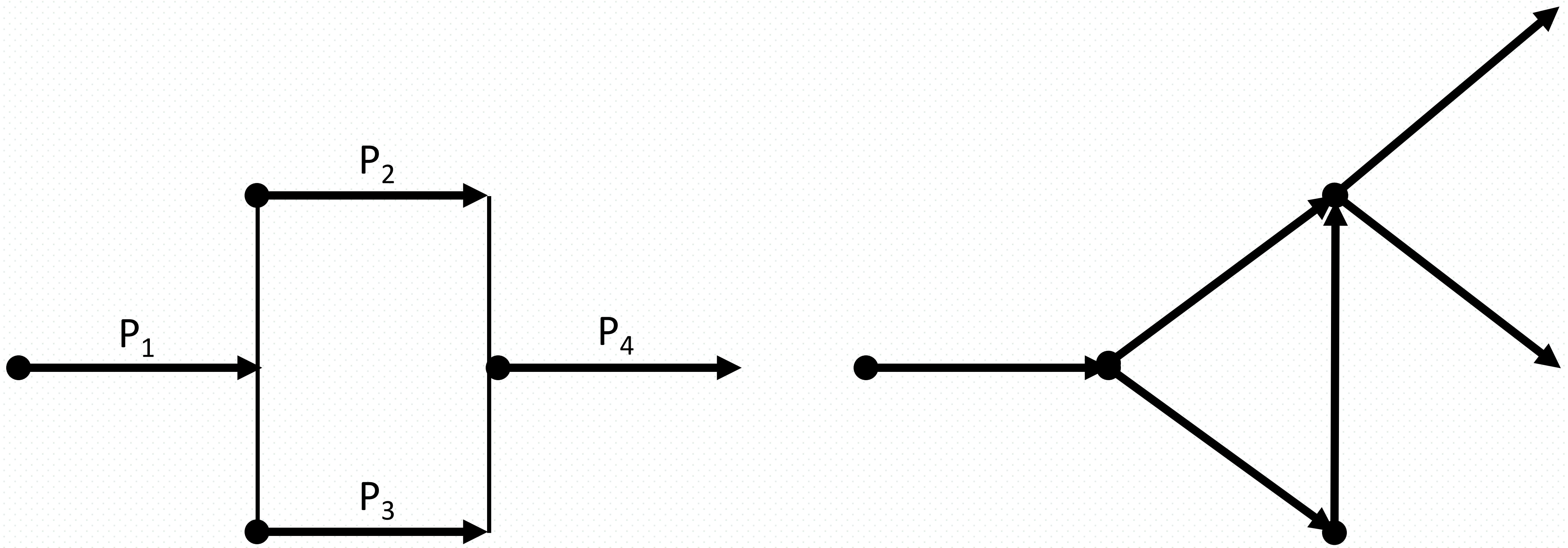
  /* S_i 代表 第[i]棒是否跑完，第[i+1]棒是否可开始，

  i= 1, 2, 3

  - initially, S1=S2=S3=0

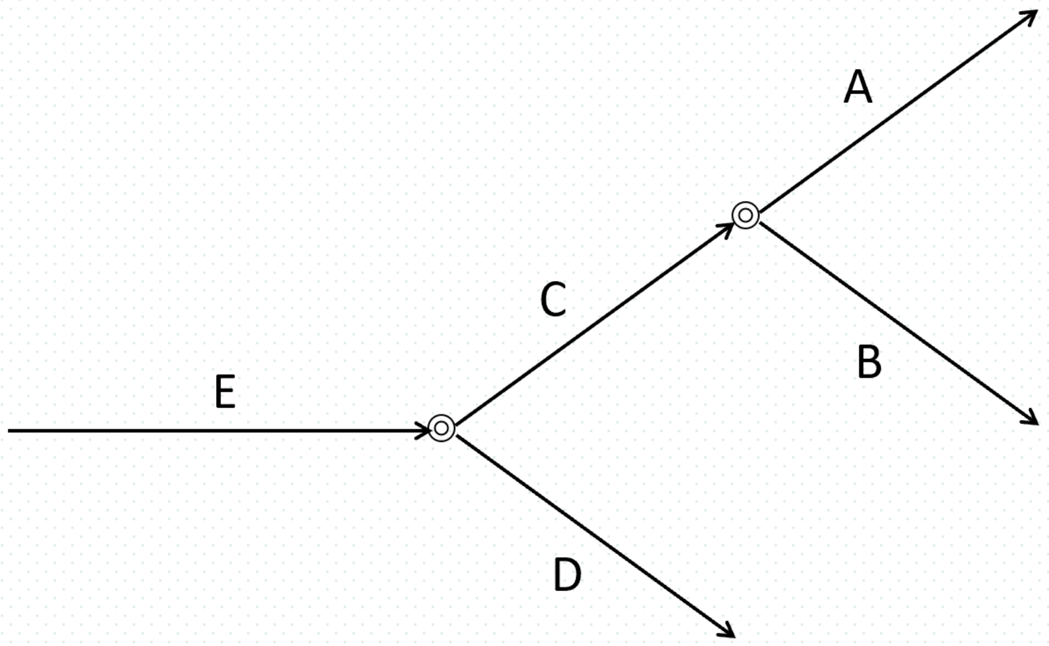
P_1:	P_2:	P_3:	P_4:
发令枪响	wait(Sl);	wait(S2);	wait(S3);
起跑，	起跑，	起跑，	起跑，
前进l00m;	前进l00m	前进l00m	前进l00m;
signal(S1);	signal(S2);	signal(S3);	到达终点

More Examples



Example 2-2

- 现有5个并发进程A、B、C、D、E，进程C完成之后A和B才开始执行，进程E完成之后C和D才开始执行
- 使用信号量和wait()、signal操作，描述上述进程间的同步关系
- 要求
  - 说明信号量含义及其初值，信号量名称应有明确含义，与题意相符
- 答案
  - 5个进程间同步关系



Example 2-2

- 本题考查进程间同步，设计如下进程间二元同步信号量：
  - Semaphores

    syncC-A=0;　//C完成后，通知A

    syncC-B=0;　//C完成后，通知B

    syncE-C=0;　//E完成后，通知C

    syncE-D=0;　//E完成后，通知D

- 说明：

1. 缺少1个信号量或信号量初值、信号量初值设置不对
2. 信号量名称含义不明，如定义为x1、x2、x3、x4

扣分!!

Example 2-2答案

A:

wait(syncC-A);
工作区
退出结束

B:

wait(syncC-B);
工作区
退出结束

C:

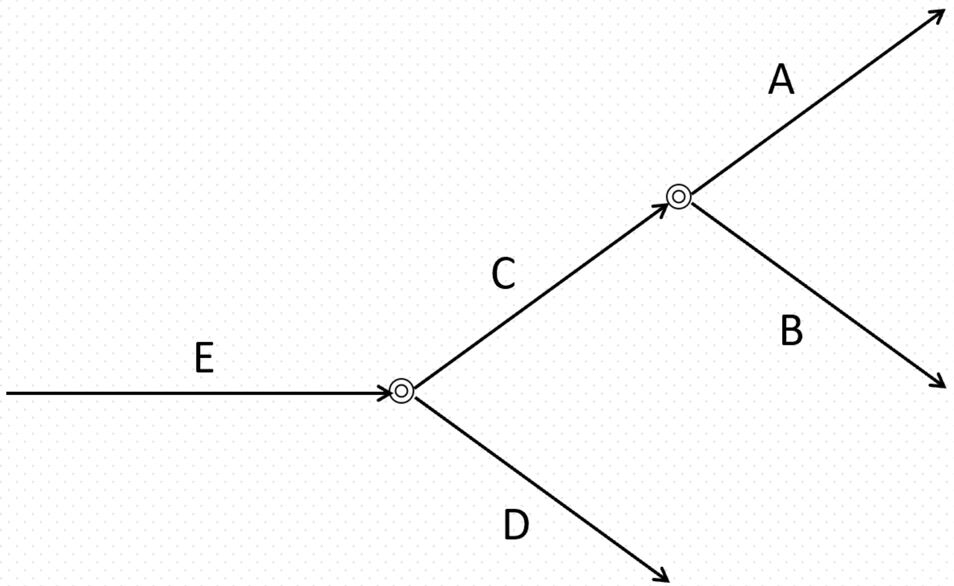
wait(syncE-C);
工作区
signal(syncC-A);
signal(syncC-B)

D:

wait(syncE-D);
工作区
退出结束

E:

工作区
signal(syncE-C);
signal(syncE-D);



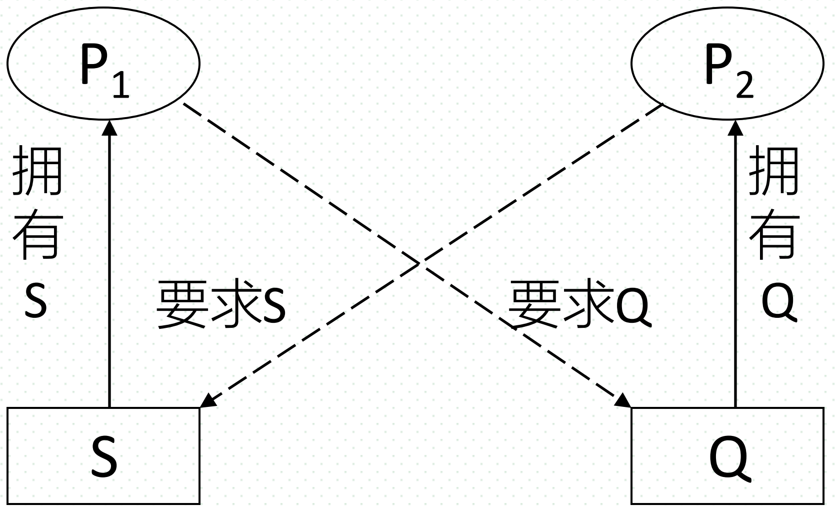
Deadlock

- **Definition**
  - states/situations related to more than one process in the system *, in which the* processes are waiting indefinitely for an event or events (e.g., *resources acquisition and release*) that can be caused by only one of the waiting processes.
  - 系统内多个并发进程彼此互相等待对方所拥有的资源，且这些并发进程在得到对方的资源之前不会释放自己所拥有的资源，造成各个进程都想得到资源而又都得不到资源，各并发进程不能继续执行的状态.

Deadlock Example

- Let *S* and *Q* be two semaphores initialized to 1

P_1	P_2
wait(S);	
	wait(Q);
wait(Q);	
	wait(S);
⋮	⋮
signal(S);	
	signal(Q);
signal(Q);	
	signal(S);



- semaphore S for resource S, semaphore Q for resource Q

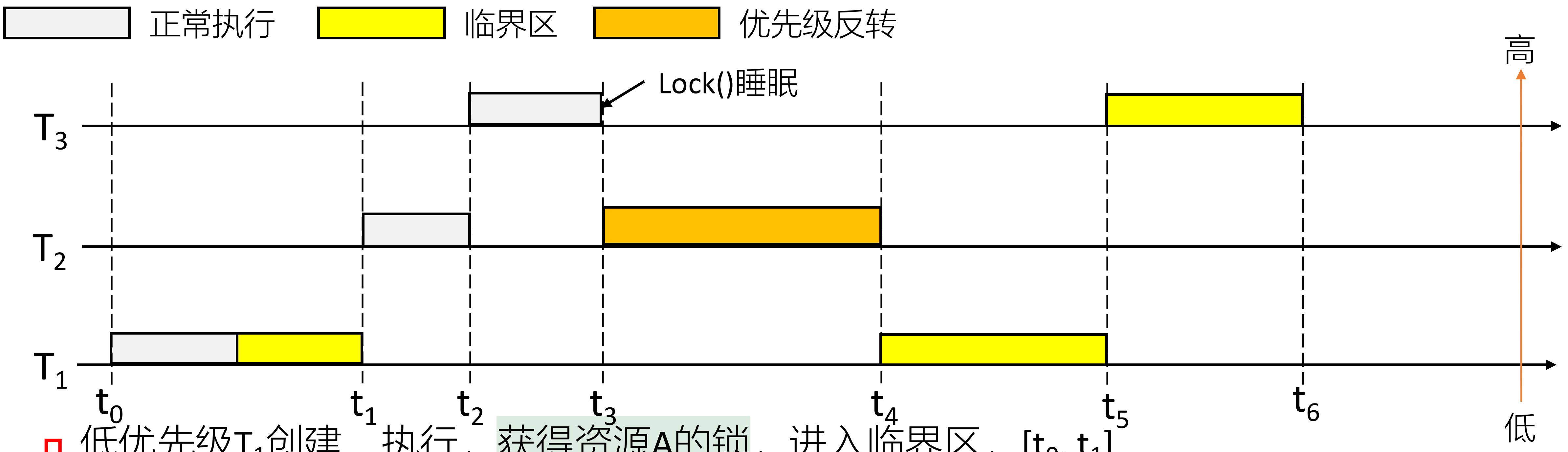
Starvation

- **Starvation**
  - ***states /situations*** related to a process (or some processes), in which the process may never be removed from the semaphore queue in which it is suspended, or a process waits indefinitely within the semaphore.
  - *infinite* waiting or blocking

- Another example of process starvation
  - the process with lower priority may be starved in priority based CPU scheduling

Priority Inversion (优先级反转)【略】

- Priority Inversion
  - 进程/线程间资源竞争引发同步互斥，导致线程占用CPU的执行顺序违反预设的优先级规定的执行顺序
    - CPU占用的顺序/优先级与资源申请/使用的顺序不一致
- 原因/示例
  - 1. 高优先级进程优先获得CPU运行
  - 2. 运行过程中，高优先级进程申请使用互斥资源，但该资源已被其它进程占用，高优先级进程被迫转入waiting态，放弃占用的CPU
  - 3. 之后，CPU被分配给低优先级进程，低优先级进程运行，先于高优先级进程执行完毕
- 实时系统中，优先级反转导致被调度的线程无法按照优先级顺序执行，某些进程/线程可能无法在deadline之前，完成任务，造成严重后果
- 假设：单核实时系统中
  - 线程T_1、T_2、T_3优先级由低到高，OS采用抢占式调度策略
  - 低优先级线程T_1与高优先级线程T_3竞争同一把锁



- 低优先级T_1创建、执行，获得资源A的锁，进入临界区，$[t_0, t_1]$
- 与T_1无资源竞争关系的中优先级T_2创建，抢占临界区中T_1的CPU，执行，$[t_1, t_2]$，T_1进入ready态
- 高优先级T_3创建，抢占T_2，执行，$[t_2, t_3)$，T_2进入ready态；t_3时刻申请加锁资源A，失败，进入睡眠等待waiting态，释放CPU
- 优先级较高的T_2获得CPU，执行，优先级反转，直至结束，$[t_3, t_4]$
- ready态下的T_1获得CPU，在临界区继续执行，直至退出临界区，释放锁并激活T_3，结束，$[t_4, t_5]$
- T_3获得锁，进入临界区中执行，直至退出临界区，释放锁，结束，$[t_4, t_5]$

Priority Inversion

- 解决方案1

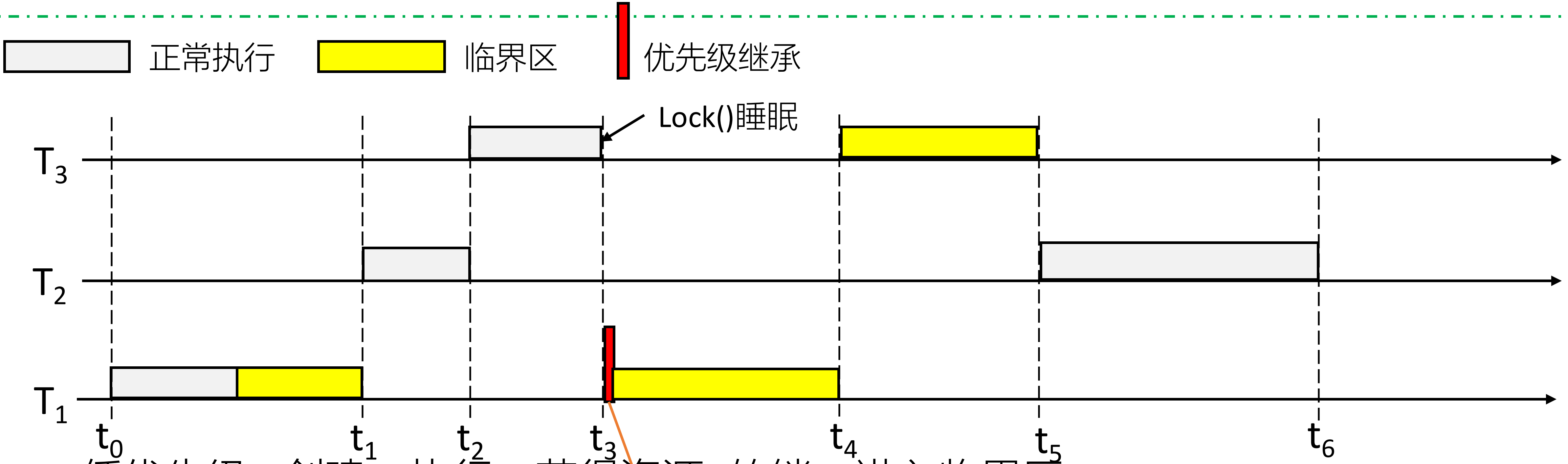
  不可抢占临界区协议(Non-preemptive Critical section Protocol, NCP)

  - 不允许线程在临界区中被抢占
  - e.g. 低优先级T_1进入临界区，$[t_0, t_1]$，不允许后创建的高优先级T_2、 T_3抢占其CPU，保证T_1可以在临界区中正常执行完

- 解决方案2

  优先级继承协议(Priority Inheritance Protocol, PIP)

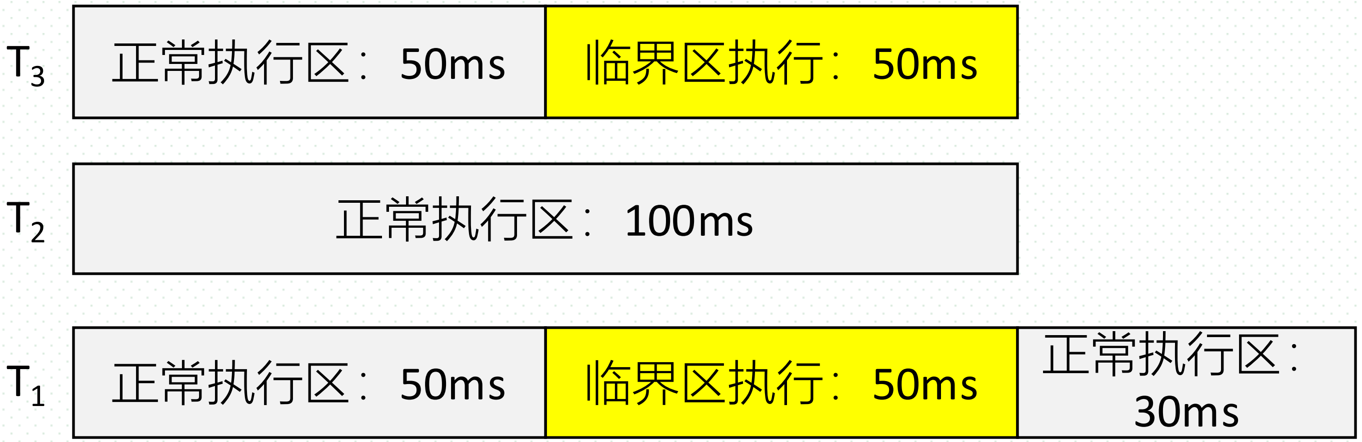
  - 当高优先级线程申请锁访问资源得不到满足，而处于阻塞态时，使锁的持有者继承其优先级，避免低优先级线程在临界区中被其它线程抢占CPU资源



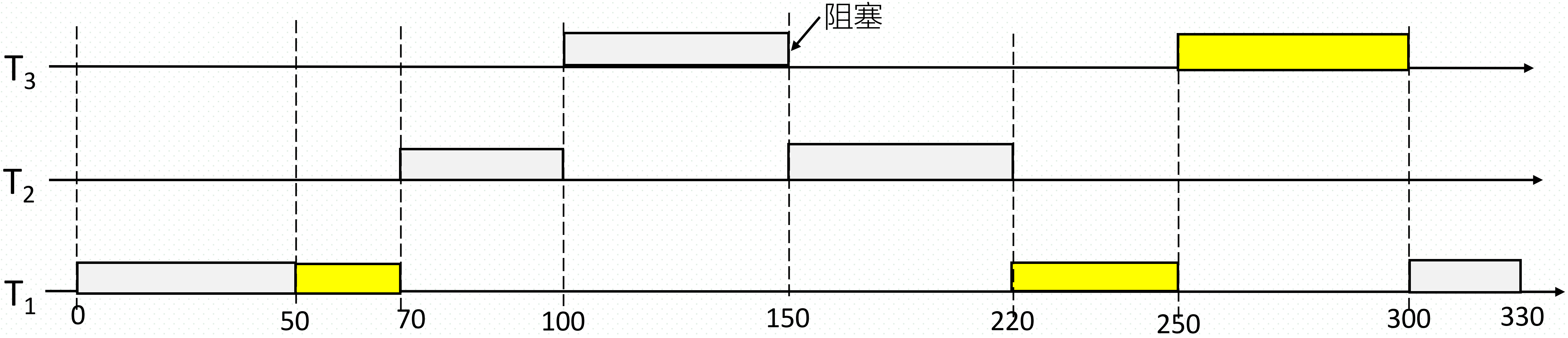
- 低优先级T_1创建、执行，获得资源A的锁，进入临界区，$[t_0, t_1]$
- 中优先级T_2创建，抢占临界区中T_1的CPU，执行，$[t_1, t_2]$， T_1进入ready态
- 高优先级T_3创建，抢占T_2，执行，$[t_2, t_3)$， T_2进入ready态； t_3时刻申请加锁，失败，进入睡眠等待waiting态，同时将高优先级传递给临界区中的T_1；
- T_1继承高优先级，从ready态进入running态，继续在临界区中执行，直至退出临界区，释放锁，激活T_3， T_3进入ready态，结束，$[t_3, t_4]$
- T_3获得锁，优先进入临界区中执行，直至退出临界区，释放锁，结束，$[t_4, t_5]$
- 与资源竞争无关的T_2获得CPU，执行，直至结束，$[t_5, t_6]$

作业/试题【略】

T1
正常执行区：50ms
临界区执行：50ms
正常执行区：30ms
T2
正常执行区：100ms
正常执行区：30ms
T3
正常执行区：50ms
临界区执行：50ms
T1
正常执行区：30ms
临界区执行：50ms
正常执行区：40ms
T2
正常执行区：80ms
T3
正常执行区：50ms
临界区执行：30ms



T_1、 T_2、 T_3进入系统的时间分别为：0ms， 70ms， 100ms

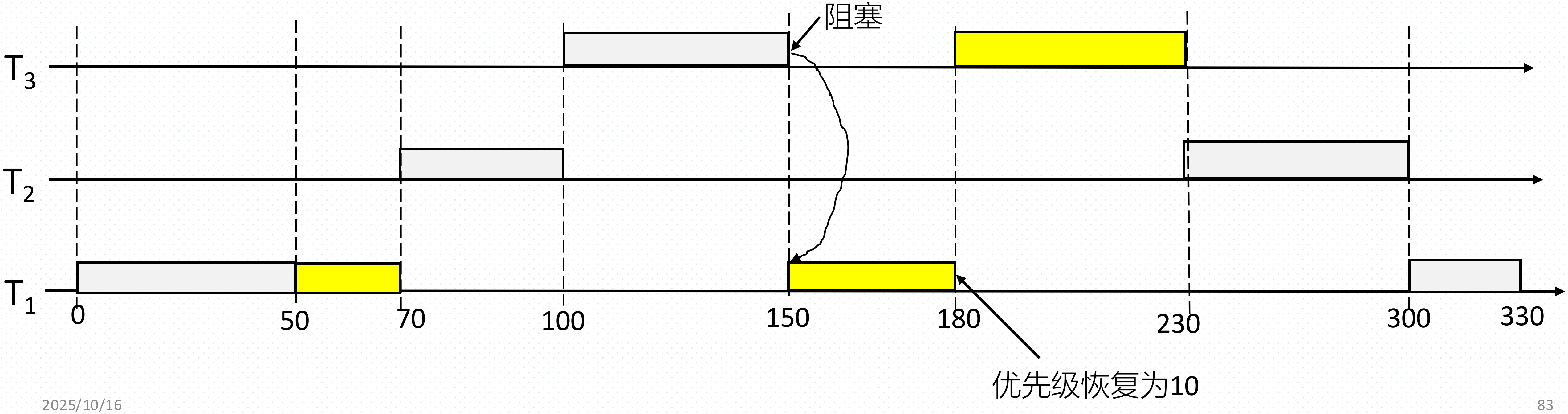


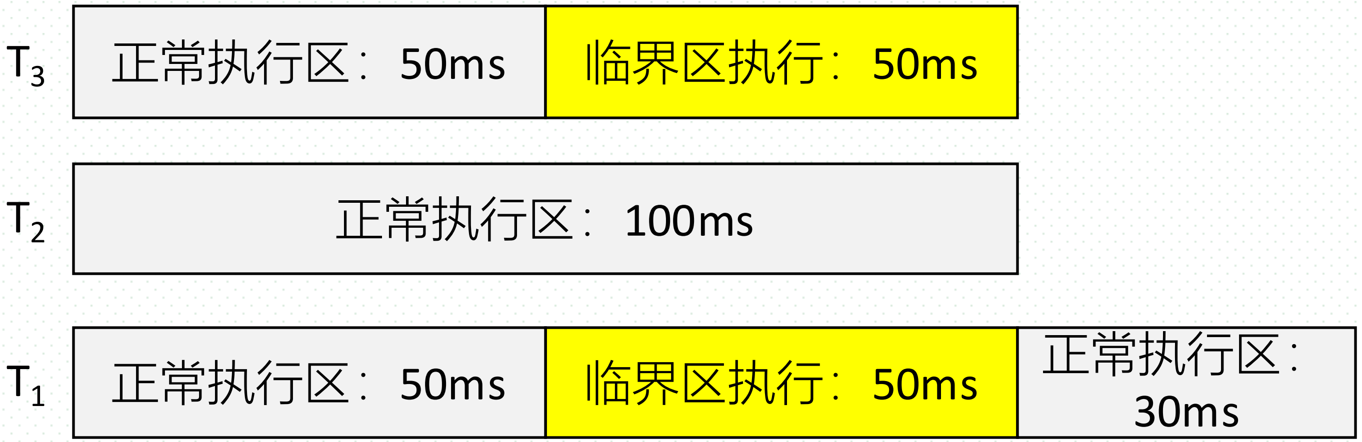
T_3	正常执行区：50ms	临界区执行：50ms	
T_2	正常执行区：100ms		
T_1	正常执行区：50ms	临界区执行：50ms	正常执行区：30ms

T_1、T_2、T_3进入系统的时间分别为：0ms，70ms，100ms

优先级继承协议：答案1

T_3在进入阻塞前，将其高优先级30传递给处于临界区中的T_1，使得T_1继续执行临界区代码；T_1退出临界区后，其优先级恢复为原有值10

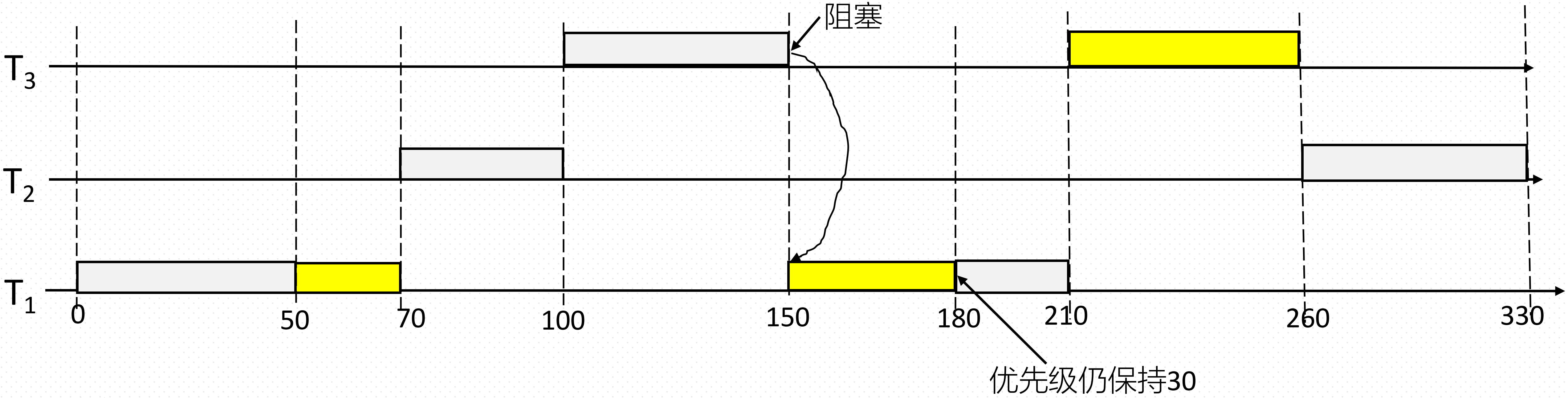


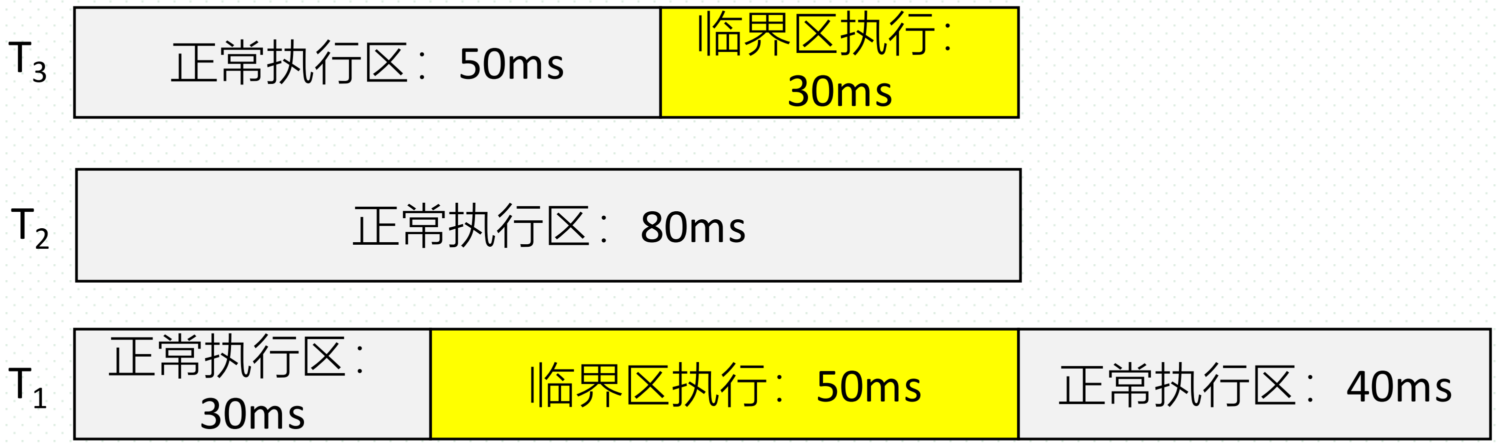


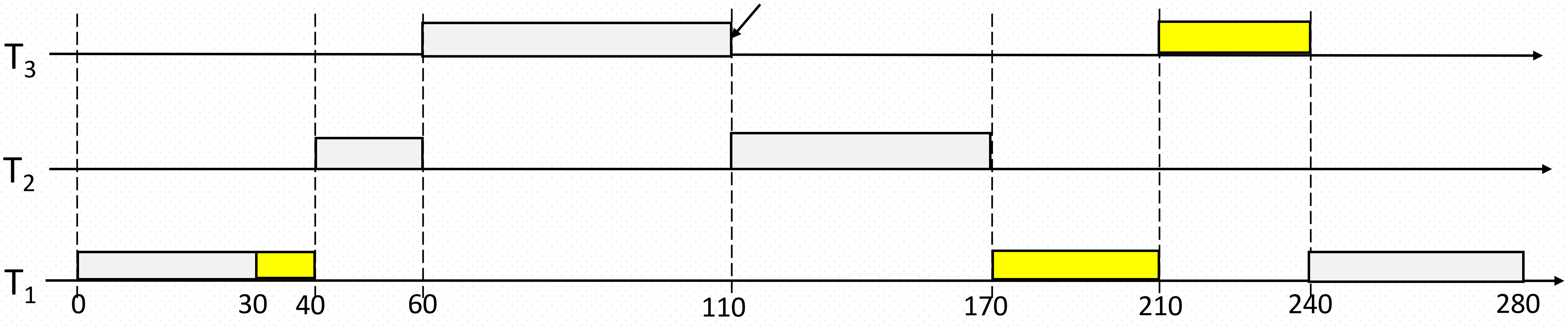
T_1、T_2、T_3进入系统的时间分别为：0ms，70ms，100ms

优先级继承协议：答案2

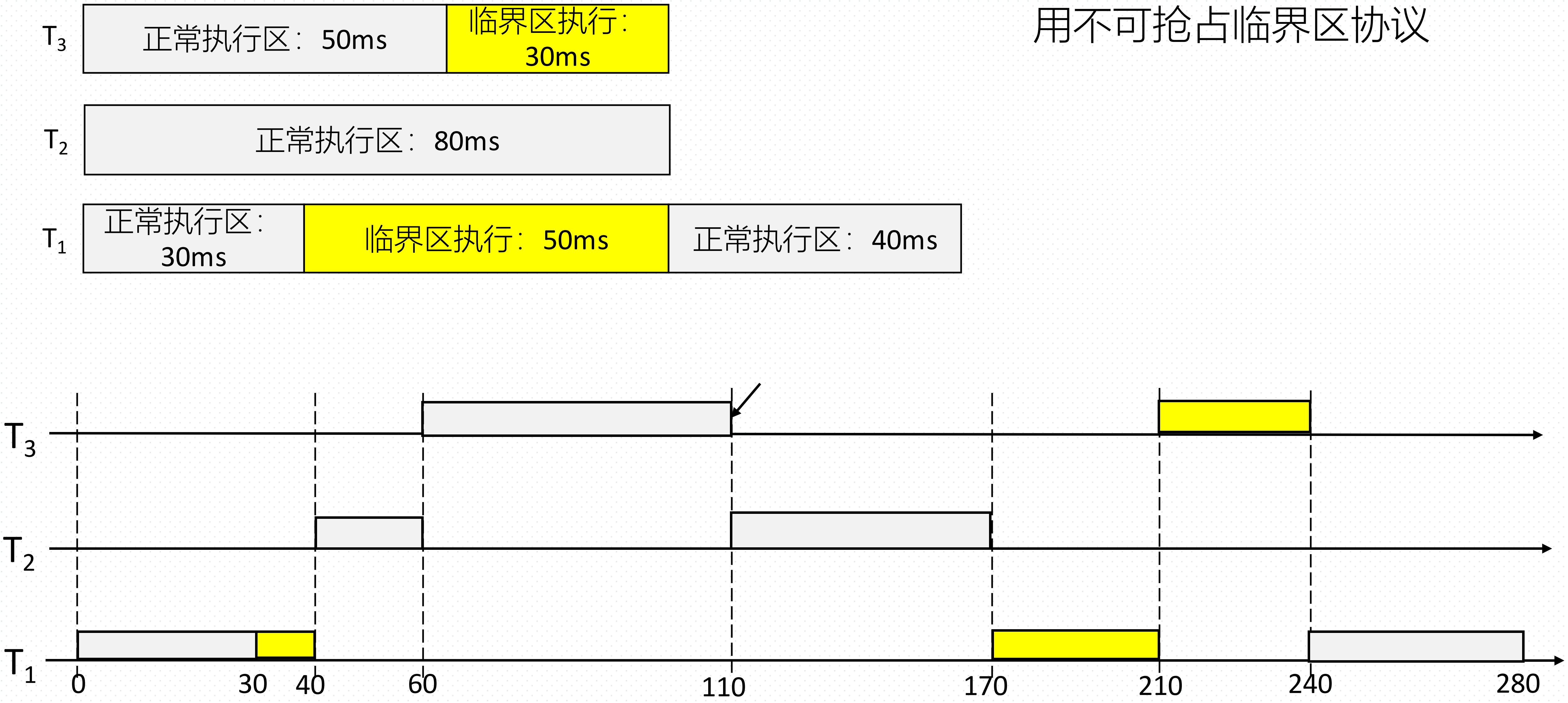
T_3在进入阻塞前，将其高优先级30传递给处于临界区中的T_1，使得T_1继续执行临界区代码，但T_1退出临界区后，其优先级仍然保持30，继续运行



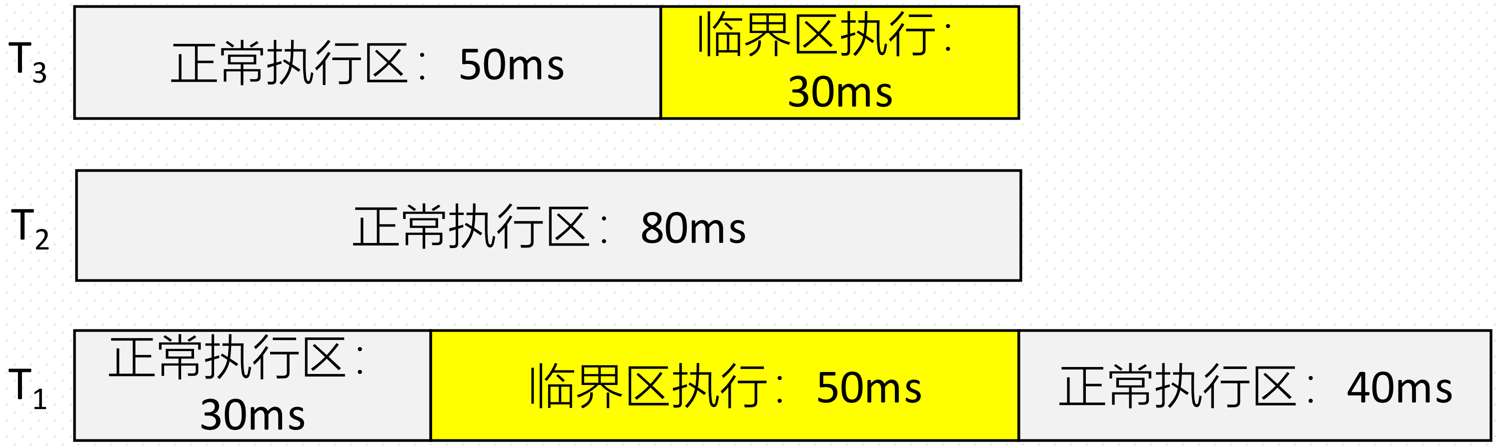

T3
正常执行区：50ms
临界区执行：30ms
T2
正常执行区：80ms
T1
正常执行区：30ms
临界区执行：50ms
正常执行区：40ms


T3
T2
T1
0
30
40
60
110
170
210
240
280

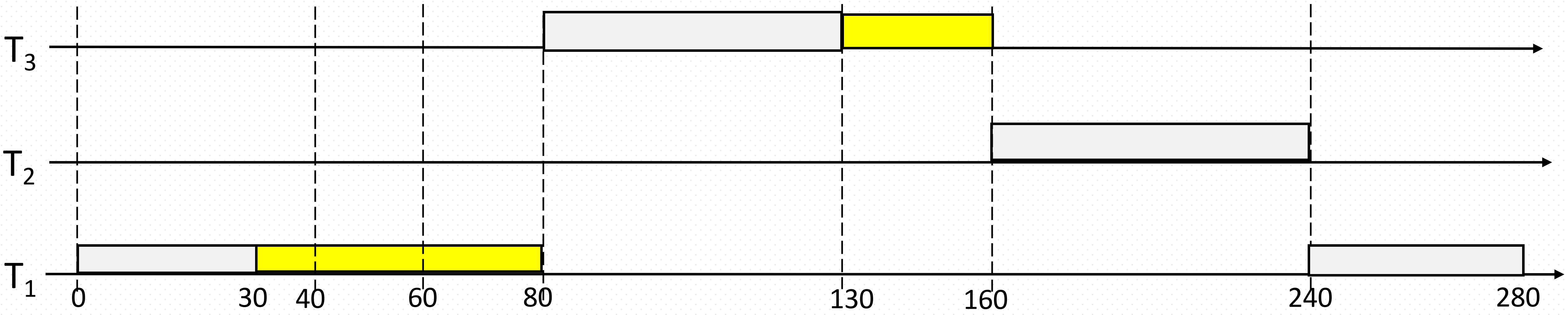
用不可抢占临界区协议



T3
正常执行区：50ms
临界区执行：30ms
T2
正常执行区：80ms
T1
正常执行区：30ms
临界区执行：50ms
正常执行区：40ms
T3
T2
T1
0
30
40
60
110
170
210
240
280



采用不可抢占临界区协议



6.6 Classical Problems of Synchronization

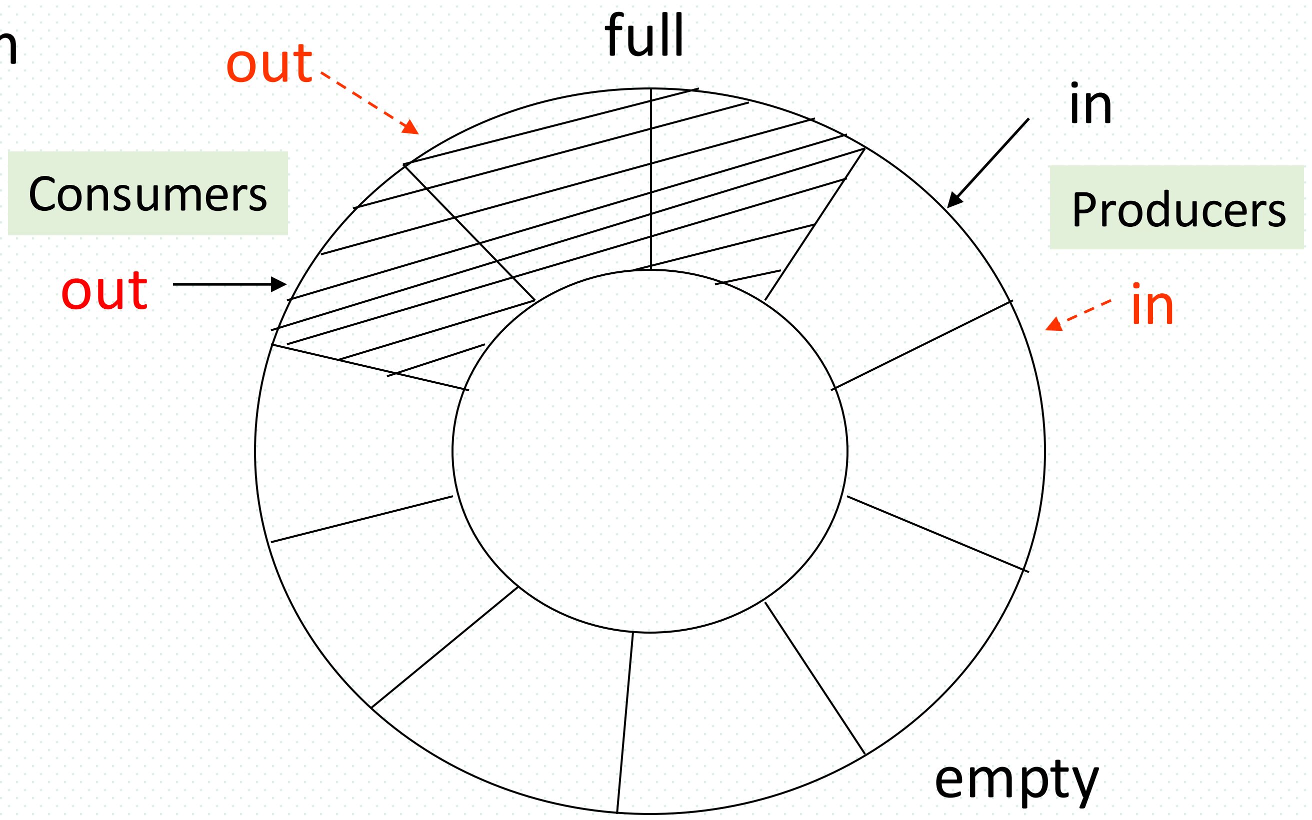
- Bounded-Buffer/Producer-Consumer Problem （有限缓冲区/生产者-消费者）
- Readers-Writers Problem（读写者）
- Dining-Philosophers Problem （哲学家就餐）
- Sleeping-Barber problem （睡眠理发师）

Bounded-Buffer Problem

- Bounded-buffer problem
- Extended bounded-buffer problem
  - 例3. Extended Bounded-Buffer Problem 1: 同时具有生产者/消费者角色
    - 老和尚-小和尚挑水，流水生产线
  - 例4.Extended Bounded-Buffer Problem 2：生产者/消费者同时进入buffer
  - 例5.Extended Bounded-Buffer Problem 3：多类产品 in buffer
  - 例6.Extended Bounded-Buffer Problem 4：
    - 每次从buffer中取出m>1个产品，或者向buffer中放入n>1个产品
  - 期末试题/实验编程题
  - ±n(n>1)的信号量wait()、signal()操作

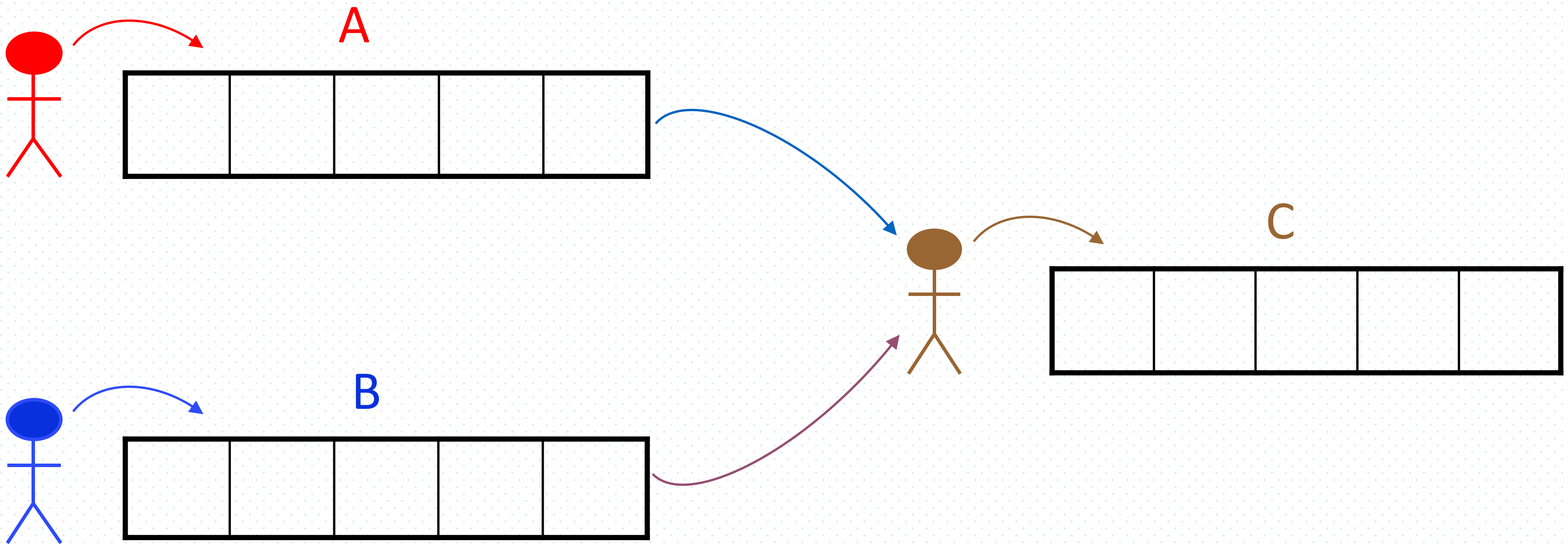
Bounded-Buffer Problem

- Also known as Producer-consumer problem
- Mutual exclusion among
  - producers
  - consumers
  - producers and consumers
- Mutual exclusion objects
  - empty units, full units in the buffer



Bounded-Buffer Problem

- E.g.1 mail-based inter-process communications
- E.g.3 A production pipeline consisting of 3 steps, i.e. A, B, and C, and each step holds a bounded buffer to store the components produced



Bounded-Buffer Problem

- **Semaphores**
  - **mutex**
    - *binary semaphore*, guaranteeing mutual exclusively operating on the buffer, i.e. only one process is allowed to operate on the buffer
    - 可扩展为：允许一个producer、一个consumer同时访问buffer中不同单元
  - **full**
    - full-buffer-units available   //可用满缓冲单元数量
    - 作为counting semaphore，计数缓冲区中产品数量
  - **empty**
    - empty-buffer-unit available //可用空缓冲单元数量

- Initially
  - **full = 0,**
  - **empty = n**
  - **mutex = 1**

Producer Process

3

```
do {
     …
     produce an item in nextp
      …
    wait(empty);      /*是否有空单元*/
    wait(mutex);      /*是否可向buffer/空单元写，或是否有其它进程在操作buffer*/
        …
    add nextp to buffer  /进入临界区，写入数据
        …
   signal(mutex);     /*允许其它进程访问*/
   signal(full);      /*满单元数加1，唤醒被阻塞的消费者*/

 } while (1);
```

producer

```
do { …
      produce an item in nextp
        …
     wait(empty);
      /*是否有空单元
     wait(mutex);
      /*是否可向buffer/空单元写
     ，或是否有其它进程在操作buffer
          …
     add nextp to buffer
          …
    signal(mutex);
      /*允许其它进程访问
    signal(full);
      /*满单元数加1，唤醒被阻塞消费者

} while (1);
```

consumer

```
do {…

     wait(full)
          /*是否有满单元
     wait(mutex);
          /*是否可从buffer/满单元读
          …
     remove an item from buffer to nextc
          …
     signal(mutex);
          /*允许其它进程访问
     signal(empty);
          /*空单元数加1，唤醒被阻塞的生产者
          …
      consume the item in nextc
          …
} while (1);
```

上机作业/期末试题

- An assembly line is to produce a product C with four part As, and three part Bs 【C = 3A + 4B】
- The worker of machining(加工) A and worker of machining B will produce two part As and one part B independently each time. Then the two part As or one part B will be moved to the station(工作台), which can hold at most 12 of part As and part Bs altogether
- Two part As must be put onto the station simultaneously
  - worker A一次生产2个A，必须一次性放入工作台
- The workers must exclusively put a part on the station or get it from the station. In addition, the worker to make C must get all part of As and Bs for one product once.
  - worker C必须一次性获得所需的3个A、4个B
- Using semaphores to coordinate the three workers who are machining part A, part B and the product C to manufacture the product without deadlock. It is required that
  - (1) definition and initial value of each semaphore, and
  - (2) the algorithm to coordinate the production process for the three workers

should be given

Cautions

- To avoid deadlock,
  - for the producer processes, the operation wait(empty) should be executed before wait(mutex)
    - 先判断有无空单元（是否有访问buffer的必要），再执行*wait*(mutex)，获取对buffer的访问权
  - for the consumer processes, the operation wait(full) should be executed befor wait(mutex)
    - 先判断有无满单元（是否有访问buffer的必要），再执行*wait*(mutex)，获取对buffer的访问权
- 如果wait操作执行顺序不当，有可能产生死锁
- In exit sections, producers/consumers release the buffer at first by signal(mutex), to permit other producers/consumers to access the buffer as early as possible

Readers-Writers Problem

- Problem description
  - readers, writers, concurrently access on shared data, e.g. files or database systems
  - constraints
    - more than one reader can simultaneously enter their critical sections to access the data item
    - when a reader is accessing the data item, the other readers are also allowed to access, but no writer is permitted

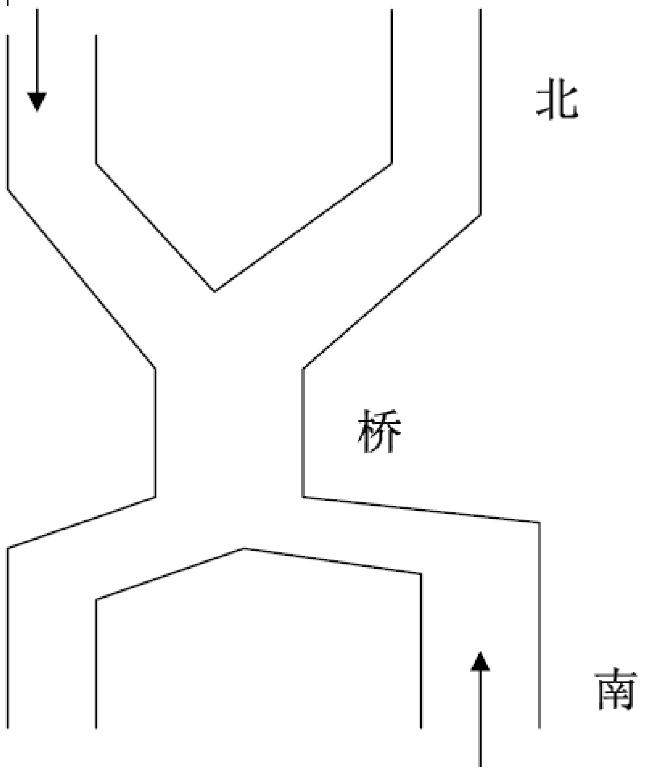
R-R	W-R	W-W
Yes	No	No

Extended Readers-Writers Problem

- E.g. Soldiers in two queues, passing a bridge
  - **two types of readers**, and each queue can be viewed as a type of readers

例题 11．(北京大学 1992 年试题)

有桥如图 4.15 所示。车流如箭头所示。桥上不允许两车交会，但允许同方向多辆车依次通行(即桥上可以有多个同方向的车)。用 P, V 操作实现交通管理以防桥上堵塞。



例. Bupt2020 软件工程考研题
（花果山猴子）
猴子作为读者+写者

Solution to Readers-Writers Problem

- Data structures
  - **semaphore wrt** /*共享数据写操作的互斥信号量
  - **Readcount** /*正在读的reader数目
  - **semaphore mutex**

    /*访问共享变量readcount的互斥信号量
- Initially **wrt = 1, mutex =1, readcount = 0**

Reader

读者优先

```
wait(mutex);            /*互斥访问readcount
readcount++;             /*读者数加1
if (readcount == 1)     /*如果是第1个读者，判断是否有写者、存在读写冲突
        wait(wrt);          互斥/禁止写操作，与写者竞争访问权
signal(mutex);          /*恢复对readcount访问*/
  ...
 reading is performed
  ...
 wait(mutex);           /*互斥访问readcount*/
 readcount--;           /*读者数减1*/
 if (readcount == 0)    /*如果已经无readers，释放数据访问权
         signal(wrt);        恢复写操作许可          */
 signal(mutex)；         /*恢复对readcount访问*/
```

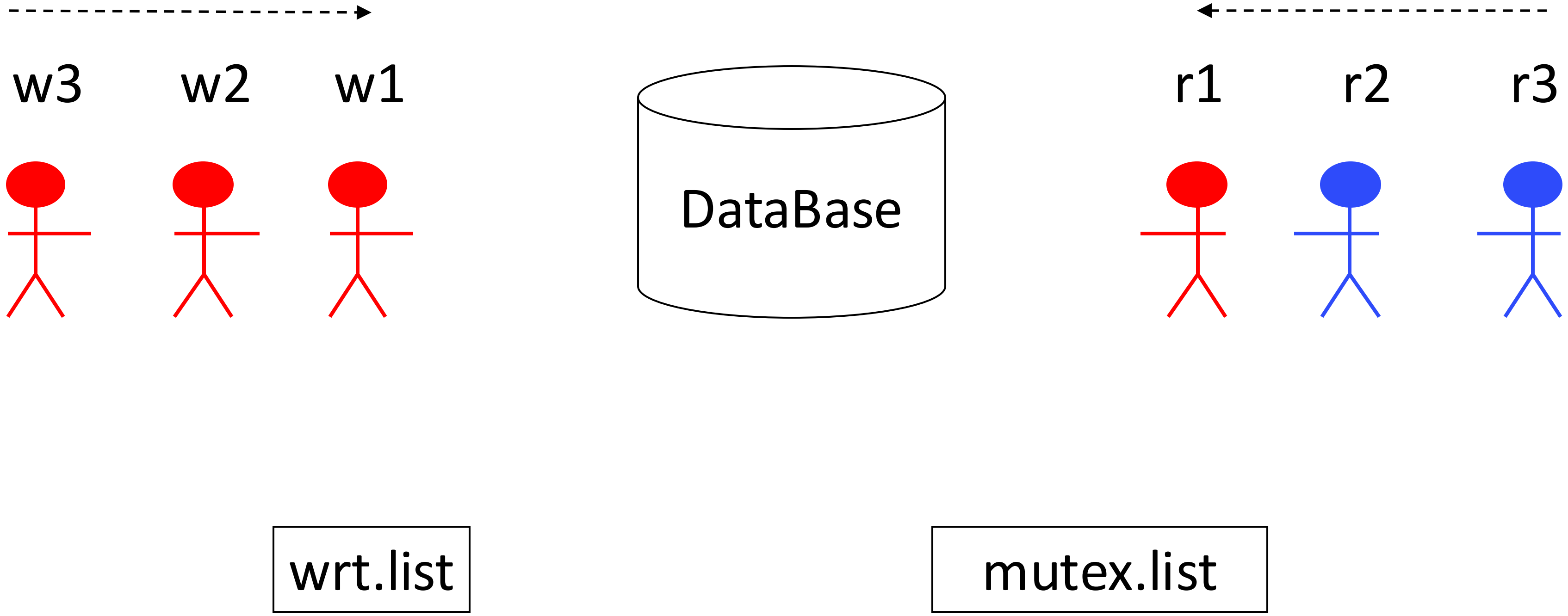
Solution to Readers-Writers Problem

- Writer Process: **wait(wrt);** **/*** 判断有无其它写者、读者正在访问

```
wait(wrt);
    …
writing is performed
    …
signal(wrt);
```

- **Note:** if a *writer* is in its critical section while *n readers* are waiting,
  - the first reader is queued on **wrt**
  - the other *n*-1 readers are queued on **mutex**

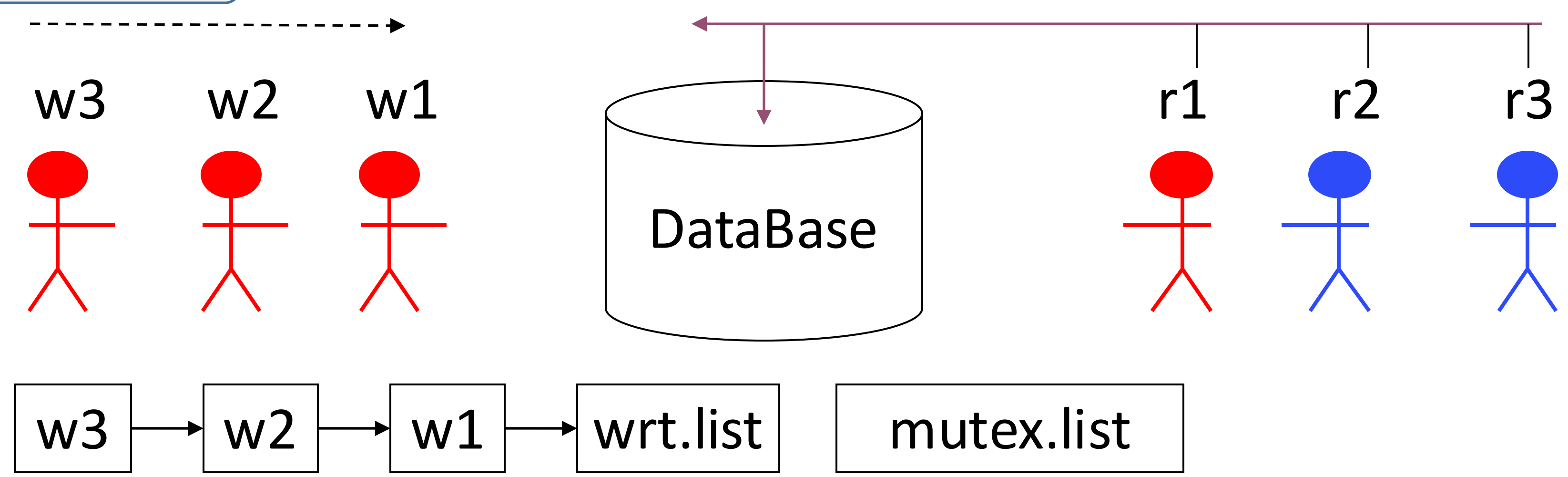
第一个读者执行wait(wrt)，负责与写者争夺控制权



initially, readcount=0, mutex=1,wrt=1

先读后写

initially, readcount=0,
mutex=1,
wrt=1



先读后写：3个读者依次进入，在临界区内并行读；当读者未完成时，稍后3个writers提出进入申请，均被阻塞：

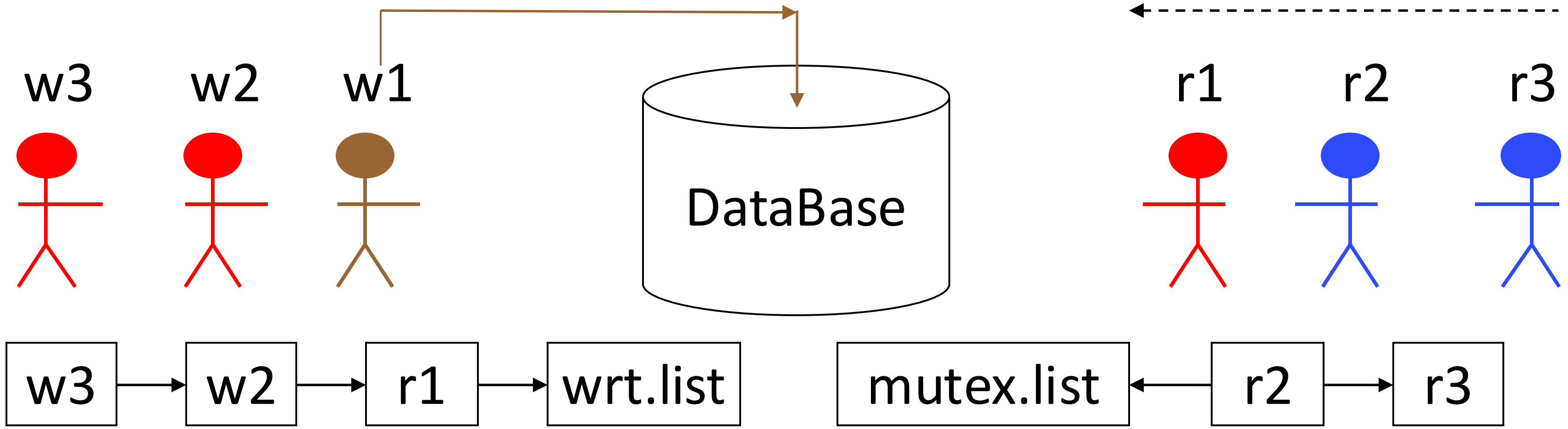
Readers	Writers	wrt	mutex	readcount
r1		1→0	1→0→1	0→1
	w1	×		
r2			1→0→1	1→2
	w2	×		
r3			1→0→1	2→3
	w3	X		

```
wait(mutex);
readcount++;
if (readcount == 1)
    wait(wrt);
signal(mutex);
```

```
wait(wrt);
…
writing is performed
signal(wrt);
```

先写后读

initially, readcount=0, mutex=1,wrt=1



先写后读：3个写者依次申请进入临界区内写；当第1个写者未完成时，稍后writers、reader提出进入申请：

Readers	Writers	wrt	mutex	readcount
	w1	1→0	1	0
r1		×	1→0	0→1
	w2	×		1
r2			×	1
	w3	×		0
r3			×	1

```
wait(mutex);
readcount++;
if (readcount == 1)
    wait(wrt);
signal(mutex);
```

```
wait(wrt);
…
writing is performed
signal(wrt);
```

Extended Readers-Writers Problems

- 例9. Extended Reader-Writer Problem 1
  - 多队读者，没有写者
  - e.g. 两队士兵相向过独木桥，两车队相向通过bridge
- 例10. Extended Reader-Writer Problem 2
  - 允许最多m个读者同时读
- 例11. Extended Reader-Writer Problem 3
  - m个读者依次读完，允许写者（防饿死）
- 例12. Extended Reader-Writer Problem 4
  - 读写者公平竞争
- 例13. Extended Reader-Writer Problem 5
  - 写者优先

Linux内核读写锁rwsem/Pthread读写锁rwlock

- Linux提供内核读写锁rwsem机制，允许更高程度的进程并行
  - 允许多个读者同时读
- 读写锁具有三种状态
  - 读模式下加锁，写模式下加锁，不加锁
- 读写锁简化编程
  - 只需要一个读写锁信号量，无需readcount、mutex等，即可实现读写者问题
- 资料：
  - Linux内核读写锁与互斥锁的区别，https://zhuanlan.zhihu.com/p/494172254?utm_source=wechat_session&utm_medium=social&utm_oi=679111968938397696&utm_campaign=shareopn
  - 剖析Linux内核《读写信号量》，https://zhuanlan.zhihu.com/p/572236863?utm_source=wechat_session&utm_medium=social&utm_oi=679111968938397696&utm_campaign=shareopn&utm_id=0

Pthread读写锁rwlock

- Linux Pthread提供读写锁rwlock（替代readcount, mutex, wrt），实现读写者问题
- 读写锁的3种状态
  - 读模式下加锁状态：允许多个读者同时占有锁，执行读操作
  - 写模式加锁状态：只允许1个写者占有锁，进行写操作
  - 不加锁状态

6个锁原语

- 初始化读写锁

  int pthread_rwlock_init(pthread_rwlock_t *rwlock, const pthread_rwlockattr_t *attr)

- 申请读锁

  int pthread_rwlock_rdlock(pthread_rwlock_t *rwlock)

- 申请写锁

  int pthread_rwlock_wrlock(pthread_rwlock_t *rwlock);

- 申请以非阻塞方式获取写锁，如果有读者或写者持有该锁，则立即失败返回，但不会被阻塞（不进入waiting态）

  int pthread_rwlock_trywrlock(pthread_rwlock_t *rwlock);

- 解锁

  int pthread_rwlock_unlock (pthread_rwlock_t *rwlock);

- 销毁读写锁

  int pthread_rwlock_destroy(pthread_rwlock_t *rwlock);

Dining-Philosophers Problem

- Problem description
  - five **philosophers** (i.e. processes), five **chopsticks** (i.e. resources), thinking and eating;
  - pick up two chopsticks that are closest to him to eat, pick up only one chopstick at a time;
  - once finish eating, put down both of her chopsticks, and start thinking
- As a model applicable in a large class of concurrency control problems
  - representing the need of **allocating several resources of same or different type among several processes** in a deadlock-free and starvation-free manner
    - e.g. multiple processes concurrently use shared resources, such as I/O devices
- 1965年，由Dijkstra提出并解决

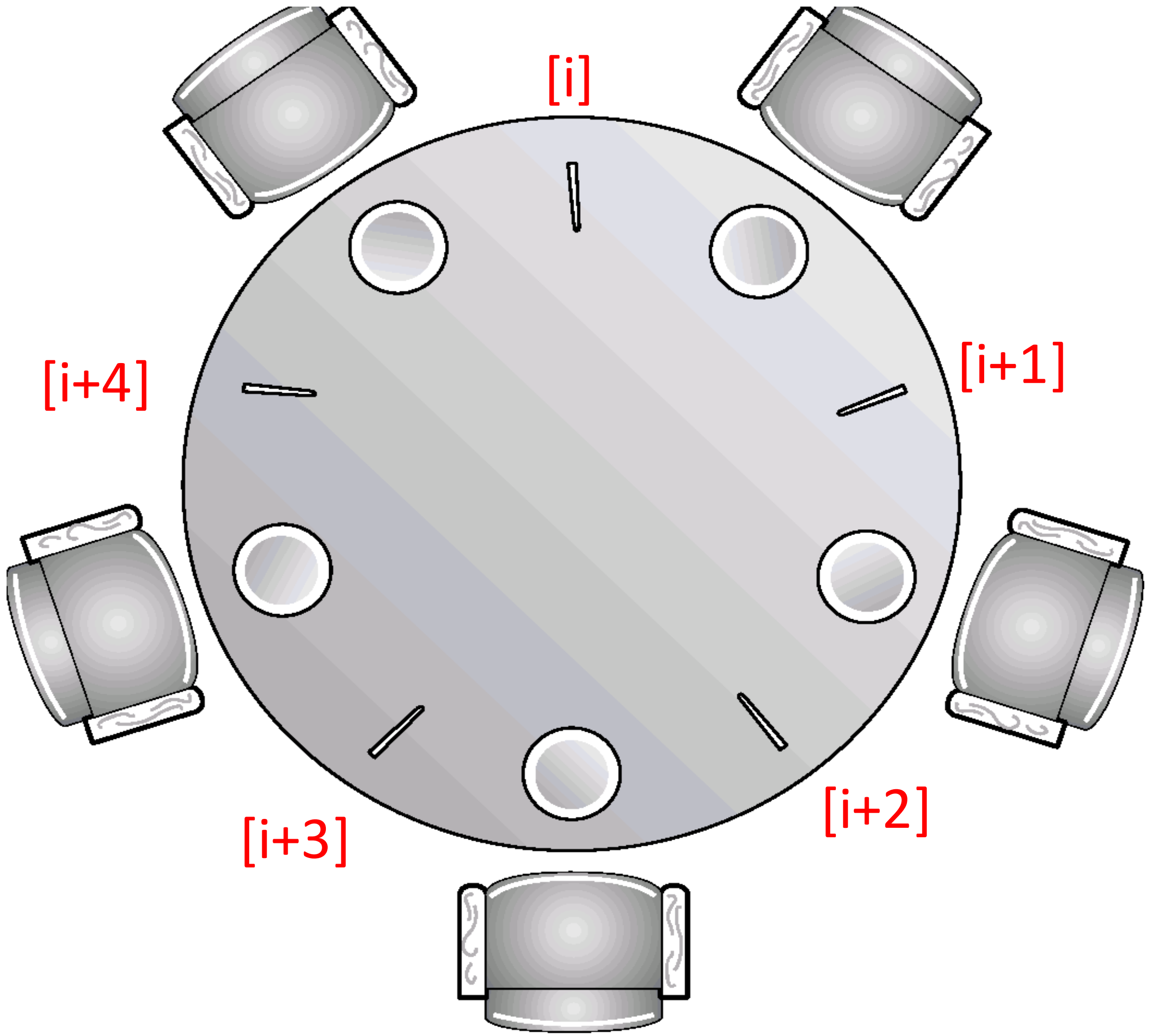


Fig. Dining-Philosophers Problem

特点:
为防止死锁，进程需要同时获得2类资源:
1) 左手刀叉 2)右手刀叉
2类资源同时申请，不允许只申请一部分资源

应用:
1. （第7章）资源预分配策略，控制死锁;
2. 课后作业:
进程执行时需要同时使用/申请1 page 内存资源和1个I/O设备

An Incorrect Solution

- An incorrect (may resulting in starvation or deadlock) solution based on semaphores
  - shared data

    semaphore chopstick[5];

    /*representing that chopstick[i] is available, $1 \le i \le 5$*/

    , initially all values are 1

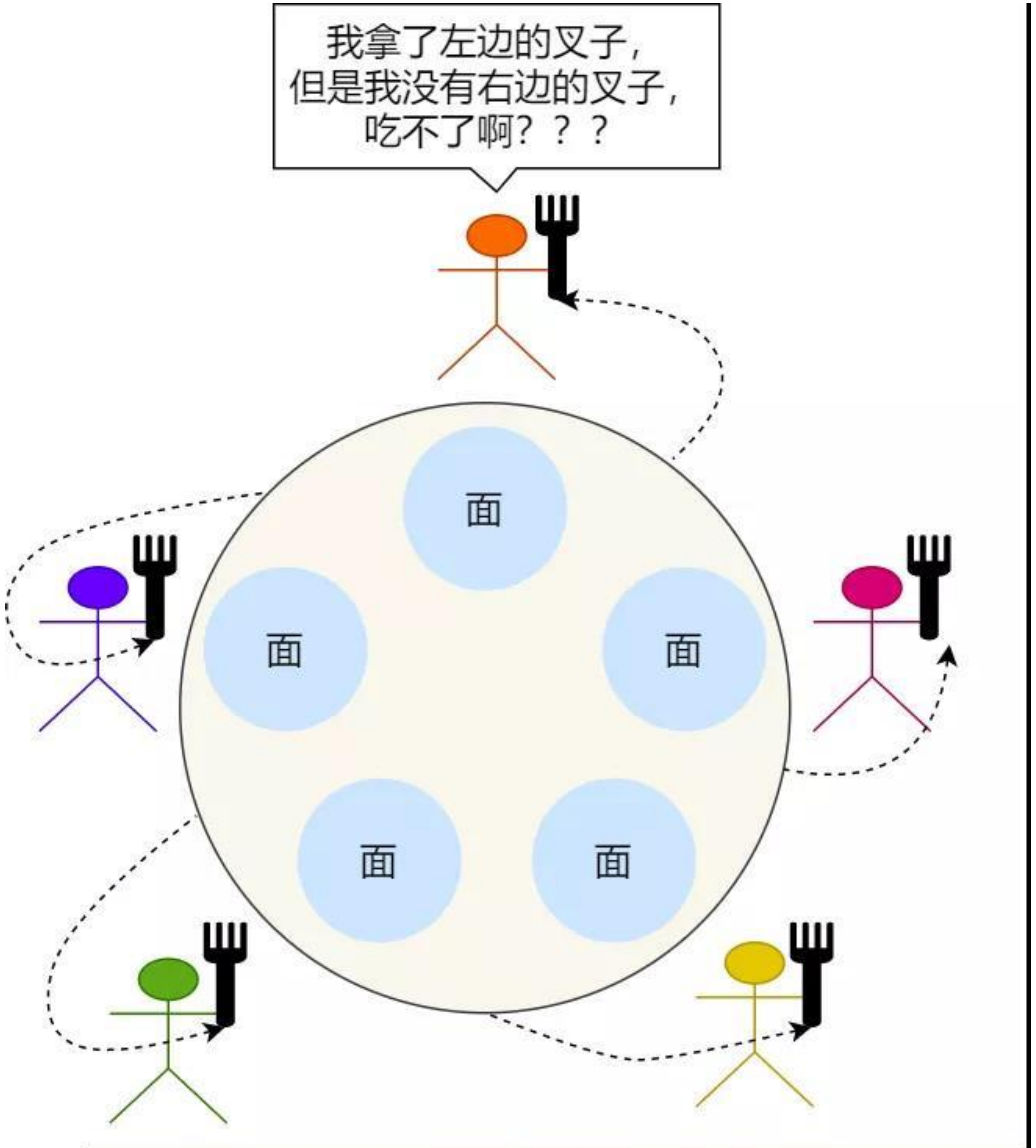
Philosopher *i*:

- Philosopher *i*:

```
do {
      wait(chopstick[i])
          /* right chopstick*/
       pickup the chopstick
      wait(chopstick[(i+1) % 5])
          /* left    chopstick*/
       pickup the chopstick
        …
      eating
        …
      signal(chopstick[i]);
      signal(chopstick[(i+1) % 5]);
        …
      think
        …
} while (1);
```

- Properties
  - guarantee that no two neighbors are eating simultaneously
  - ***deadlock*** occurs, e.g., when each philosopher picks up her left chopsticks

- 方案1及其错误



方案一的问题：
每个哲学家同时拿了左边的叉子，就会导致死锁的现象

Dining-Philosophers Problem

- Dining-Philosophers problem can also be resolved by monitors as shown in (§6.7)
- 解法1：每次最多只允许4个哲学家同时取刀叉，这样至少保证1人可拿到左右手刀叉，避免死锁
  - 设置多元信号量limit=4，每人取刀叉前，wait(limit)；取到2把刀叉后，signal(limit)
  - 问题：不适用哲学家人数动态变换的情况
- 解法2： 有序申请左右手刀叉
  - 给5把刀叉编号
  - 每人遵循死锁预防原理，按照资源有序申请原则，先申请左右手中编号较小的刀叉，再申请另一侧编号较大的刀叉
- 解法3：更为灵活、通用，不限制哲学家人数和进餐行为
  - a correct semaphore-based solution to this problem，similar to that in §6.7, is given

剩余段
进入段
临界段

- **A correct semaphore-based solution**
  - **enum {thinking, hungry, eating} state[5];**

    /*记录每个人状态*/
  - **semaphore mutex=1;**

    /*临界区互斥*，保证每时刻最多只能有一个人执行从餐桌上取或放2把叉子的动作/
  - **semaphore self[5];**

    /*哲学家信号量，表明i是否已获得2把刀叉并可以开始进餐*/
  - **initially, mutex=1, state[i]=thinking, self[i]=0**

- void philosopher(int i) {

```
while (true) {
    thinking();      /*剩余段：思考*/
    pickup(i);       /*进入段：同时获取两把叉子，或阻塞*/
    eating();        /*临界段：进餐*/
    putdown(i);      /*退出段：放下两把叉子*/

};
```

```
void pickup(int i) {
    wait(mutex);
      /*申请进入临界区访问餐桌
    state[i] = hungry;
      /*记录为饥饿
    test[i];
      /*测试左右刀叉是否可用
    signal(mutex);
      /*退出访问餐桌区
    wait(self[i])
      /* 等待左右空闲刀叉,
}        如果得不到叉子则被阻塞,
```

```
void test(int i)
{
    if ( (state[(i + 4) % 5] != eating) &&
         (state[i] == hungry) &&
         (state[(i + 1) % 5] != eating))
    {
         state[i] = eating;
         signal(self[i]);   /*唤醒i, 允许其进餐*/
    }
}
```

当左右邻居空闲、左右刀叉均可用时，self[i] 0→1

```
void putdown(int i) {
        wait(mutex);          /*申请访问餐桌*/
        state[i] = thinking;  /*记录为思考*/
         test[(i+1)%5];      /*释放左手刀叉，测试左邻居是否想进餐*/
         test[(i+4)%5];      /*释放右手刀叉，测试右邻居是否想进餐*/
        signal (mutex);       /*退出访问餐桌*/
}
```

signal(self[i+1]%5)

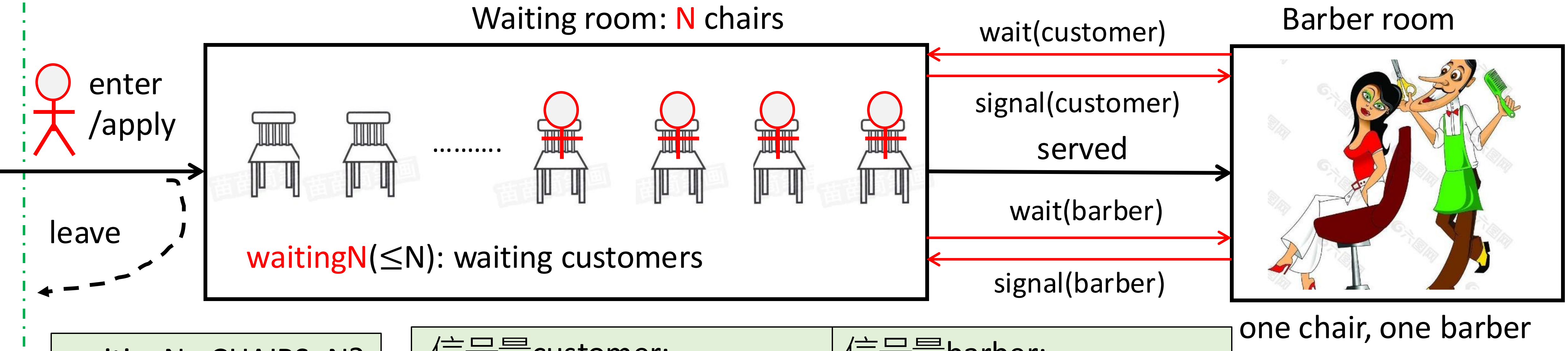
signal(self[i+4]%5)

i进餐完毕，唤醒可能被阻塞在self对列的左右邻居

Sleeping Barber Problem

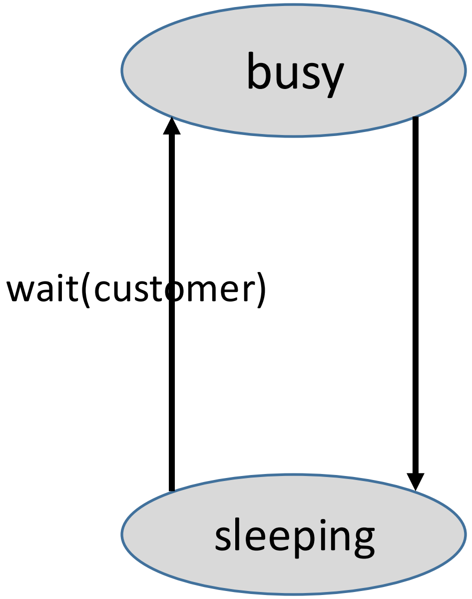
- Sleeping Barber Problem
  - A barbershop consists of a waiting room with N chairs and a barber room with one barber chair
  - If there are no customers to be severed, the barber goes to sleep
  - If a customer enters to the barbershop, and all chairs are occupied, then the customer leaves the shop
    - 此时，customer不会因为chairs资源不够用，被阻塞等待，而是离开理发店
  - If the barber is busy but the chairs are available, the customer sits in one of free chairs. If the barber is asleep, the customer wakes up the barber
- 应用
  - 作业：医院挂号看病
  - 助教答疑

customer, barber相互等

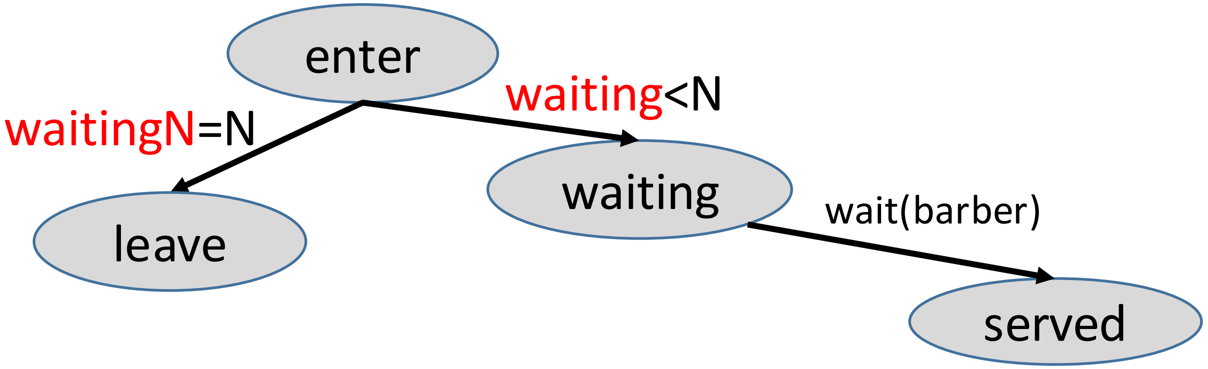


waitingN< CHAIRS=N?
决定是否进入或离开

信号量customer:	信号量barber:
customer-barber间通知是否有顾客，用于customer唤醒barber为顾客理发	barber-customer间通知barber是否空闲，用barber于唤醒等待的顾客
customer: signal(customer)	barber: signal(barber)
barber: wait(customer)	customer: wait(barber)



要求:
waiting room容量为N，当容量已满时（n个顾客等待），后续顾客离开，不能被阻塞，故引入计数变量waitingN



Sleeping Barber Problem

- Problem
  - barbershop: waiting room, barber room
  - n chairs in waiting room, one barber chair in barber room
  - customers: enter—wait or leave—severed
  - barber: sleep—be waken up—be busy/working
  - 特点
    - 进程间同步 + 利用"用户空间计数变量 + 内核空间互斥信号量"判断buffer容量、顾客数量

特点：
1. 资源竞争
   顾客竞争chairs，但不会被阻塞
   顾客竞争barbers, 可能被阻塞
2. 双向同步: 利用信号量barber, customer
   customer ↔ barber

The Sleeping Teaching Assistant

- Similar to The Sleeping Teaching Assistant in P301 in the textbook

Programming Projects

Project 1—The Sleeping Teaching Assistant

A university computer science department has a teaching assistant (TA) who helps undergraduate students with their programming assignments during regular office hours. The TA's office is rather small and has room for only one desk with a chair and computer. There are three chairs in the hallway outside the office where students can sit and wait if the TA is currently helping another student. When there are no students who need help during office hours, the TA sits at the desk and takes a nap. If a student arrives during office hours and finds the TA sleeping, the student must awaken the TA to ask for help. If a student arrives and finds the TA currently helping another student, the student sits on one of the chairs in the hallway and waits. If no chairs are available, the student will come back at a later time.

Using POSIX threads, mutex locks, and semaphores, implement a solution that coordinates the activities of the TA and the students. Details for this assignment are provided below.

- binary semaphore **customer**
  - customer-barber间通知理发师是否有顾客，customer唤醒barber为顾客理发
    customer: signal(customer)
    barber: wait(customer)
- binary semaphore **barber**
  - barber-customer间的同步/唤醒信号量

barber呼叫1个customer，进入barbe room，为其理发
barber: signal(barber)
customer: wait(barber)

- int **waitingN**
  - the number of waiting customers, a copy of semaphore customers
- binary semaphore **mutex**
  - for mutual exclusion on the variable waitingN

1. 引入waitingN统计等待的顾客总数，以便与等候室内椅子总数N进行比较，判断有无空椅子
2. 引入mutex，控制互斥访问waitingN

- Initially
  - customers=0, barber=0, mutex=1
  - waitingN=0, CHAIRS=N /*等候室椅子总数

Barber

- Void Barber(void) {

```
while (true)  {
    wait(customers);      /*如果无等待顾客，则睡觉
    wait(mutex);          /*服务顾客，将等待顾客数减1
    waitingN:=waitingN-1;
    signal(mutex);
     signal(barber);      /*理发师召唤1个等待顾客，开始理发
     cut-hair()    }
  }
```

barber工作前提：有等待顾客，wait(customers)

Customer

- Void Customers(void) {

```
        wait(mutex);
        if  (waitingN < CHAIRS) {    /*判断是否有空椅子，等待
            waitingN:= waitingN+1;
            signal(customers);    /*通知理发师，有等待顾客
            signal(mutex);
            wait(barber);            /*请求理发师为自己服务*/
            get-hair()
        }
        else  /*如果客满无空椅子，离开
         { signal(mutex)
          }
    }
```

顾客被服务前提：
1. 有空椅子，可以等待，waitingN<CHAIRS=N；
2. 被理发师召唤, wait(barber)

6.8 Monitors

- Monitor/管程
  - a type of high-level synchronization construct (i.e. not in kernel space) that allows the safe sharing of an abstract data type among concurrent processes
  - provided by programming environments （such as Concurrent C, C++, Java, C#)
- Why the monitor needed ?
  - process synchronization on the basis of semaphores is somewhat difficult for the application programmers
  - disorders of the *wait* and *signal* operations in application programs may result in **deadlock**
  - to avoid this demerit, the critical sections in programs are controlled by a centralized controller, i.e. the monitor, provided by programming languages or middleware

说明：

1. 实际编程环境/语言中，monitor具体实现机制，如C/C++中的条件变量+互斥信号量，与textbook所述并不完全一致
2. 结合具体编程环境/语言中的条件变量样例，理解管程

Monitors

- **管程**"定义
  - 由程序设计语言/编程环境提供的一种进程同步互斥机制
  - 将分散在应用进程中互斥访问临界/共享资源的临界区代码集中到管程内部，统一地管理
  - 管程可以控制共享资源，为访问这些共享资源提供一种互斥机制
  - 形式上，管程 = 局部于管程内部的公共共享变量 + 访问这些变量的过程
    - 公共共享变量代表共享资源，过程对应于临界区代码

- 在程序设计语言/编程环境，e.g. Java, C++中，monitor的具体实现形式是**条件变量 conditional variables**

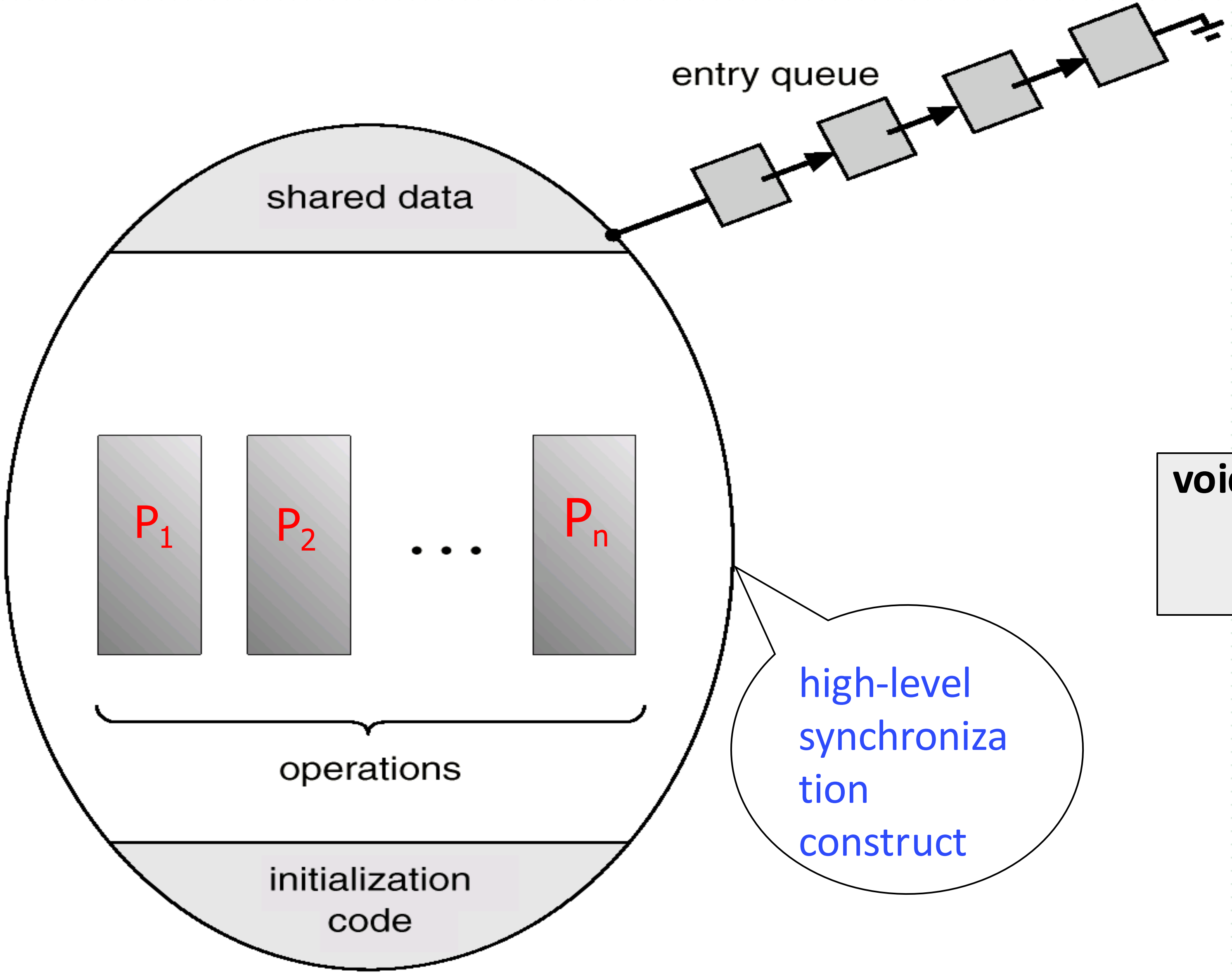
Monitors Representation

```
monitor monitor-name
{
    shared variable declarations
    procedure body P1 (…) {
                ...
    }
    procedure body P2 (…) {
        ...
    }
    procedure body Pn (…) {
        ...
    }
    {
        initialization code
    }
}
```

- Procedure $P_1, P_2, \ldots, P_n$ operate on shared variables declared within the monitor
- Monitor consists of
  - shared variables
  - concurrent procedures, e.g. *producers*, *consumers*
  - initialization codes
- Monitor construct ensures that *only one process at a time* can be active within the monitor (access the shared variables by calling P_1, P_2, …, P_n)
  - act as mutual exclusive semaphore *mutex*

  // 不需要用户程序显示地定义、使用 mutex信号量

Fig. Schematic view of a monitor



monitor pc //*producer-consumer*
{
shared variable buffer
void add(int i)
void remove(int i)
initialization code
}

void add(int i) { add **nextp** to buffer **}**	**void remove(int i)** { remove from buffer **}**

替代了对buffer的**mutex**

Producer i:
wait(empty)
pc.add
signal(full)

Consumer i:
wait(full)
pc.remove
signal(empty)

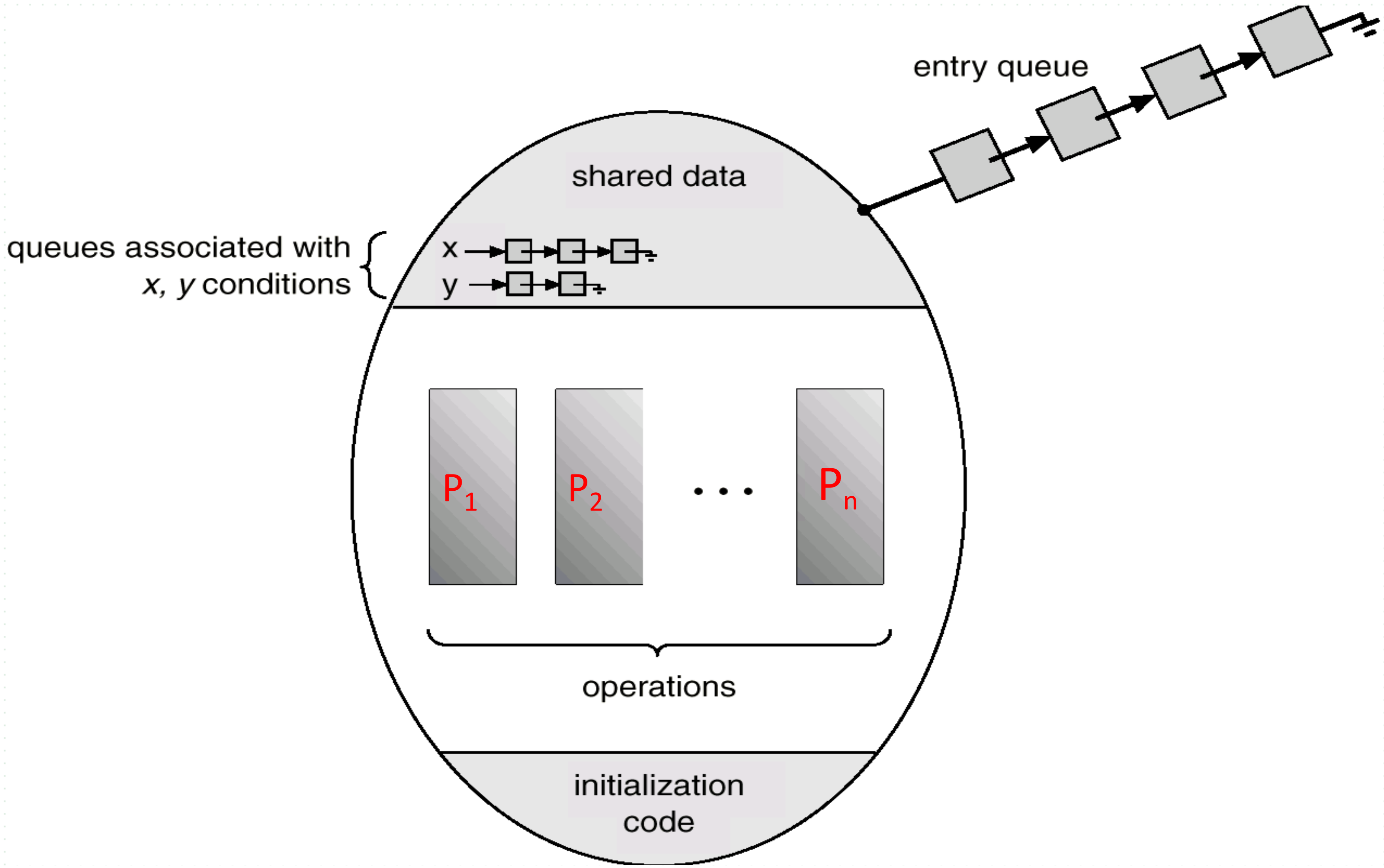
Monitor

- Improvement
  - to refining the synchronizing ability of the monitor, a process can be allowed to wait within the monitor, so condition variables 【条件变量，二元】 are introduced into the monitor, which are declared, as:

    condition x, y

- Monitor consists of
  - shared variables
  - conditional variables【条件变量】
  - concurrent procedures
  - initialization codes
- Fig. 6.18

Fig. Monitor With Condition Variables【条件变量】



Monitor

- Condition variable can only be used with the operations **wait** and **signal**. These two operators are in procedures P_1, P_2, …, P_n
  - **x.wait()**
    - means that the process invoking this operation is suspended until another process invokes
  - **x.signal()**
    - the **x.signal** operation resumes exactly one suspended process. If no process is suspended, then the **signal** operation has no effect

Bounded-buffer Monitor

```
monitor pc
 {
        shared variable buffer
        condition empty, full
        void add(int i)
        void remove(int i)

        void init() {
              empty=n;
              full=0
          }
   }
```

```
void add(int i) {
       empty.wait
       access and add nextp to buffer
       full.signal
       }


void remove(int i) {
       full.wait
       access and remove from buffer
       to nextc
       empty.signal
       }
```

```
Producer i:
       pc.add(i)

Consumer i:
       pc.remove(i)
```

问题：
　条件变量属于二值变量，如何计数空、满单元数目？
方法：引入整型变量，替代empty、full的计数功能
参见：基于条件变量和互斥锁的生产者-消费者问题

Monitor for Dining Philosophers

与信号量解法相比，无需
pickup、putdown中的mutex

- Monitor **dp**

```
{
1.  enum {thinking, hungry, eating} state[5];  ----- shared variables
2.  condition self[5];                          -----conditional variables
3.  void pickup(int i)
    void putdown(int i)                          concurrent procedures
    void test(int i)

4. void init() {
        for (int i = 0; i < 5; i++)
                state[i] = thinking;             initialization code
   }
}
```

Monitor for Dining Philosophers

```
void pickup(int i) {
      state[i] = hungry;
      test[i];
      if (state[i] != eating)
            self[i].wait();
 }
```

```
void putdown(int i) {
      state[i] = thinking;
      /* test left and right neighbors
      test((i+4) % 5);
      test((i+1) % 5);
}
```

```
void test(int i) {
if ( (state[(i + 4) % 5] != eating) &&
        (state[i] == hungry) &&
        (state[(i + 1) % 5] != eating)) {
              state[i] = eating;
              self[i].signal();
        }
 }
```

```
Philosopher i:
        dp.pickup (i)
        eat
        dp.putdown(i)
```

Java Monitor

JAVA MONITORS

Java provides a monitor-like concurrency mechanism for thread synchronization. Every object in Java has associated with it a single lock. When a method is declared to be `synchronized`, calling the method requires owning the lock for the object. We declare a `synchronized` method by placing the `synchronized` keyword in the method definition. The following defines `safeMethod()` as `synchronized`, for example:

```
public class SimpleClass {
   . . .
   public synchronized void safeMethod() {
    . . .
    /* Implementation of safeMethod() */
    . . .
   }
}
```

Next, we create an object instance of `SimpleClass`, such as the following:

```
SimpleClass sc = new SimpleClass();
```

Invoking `sc.safeMethod()` method requires owning the lock on the object instance `sc`. If the lock is already owned by another thread, the thread calling the `synchronized` method blocks and is placed in the ***entry set*** for the object's lock. The entry set represents the set of threads waiting for the lock to become available. If the lock is available when a `synchronized` method is called, the calling thread becomes the owner of the object's lock and can enter the method. The lock is released when the thread exits the method. A thread from the entry set is then selected as the new owner of the lock.

Java also provides `wait()` and `notify()` methods, which are similar in function to the `wait()` and `signal()` statements for a monitor. The Java API provides support for semaphores, condition variables, and mutex locks (among other concurrency mechanisms) in the `java.util.concurrent` package.

基于条件变量和互斥锁的生产者-消费者问题【C/C++程序】

C/C++等高级程序设计语言提供了条件变量、锁等同步互斥机制

```
int  empty=5;
int  full=0;
struc  cond  empty_cond;
lock    empty_cnt_lock;
struc  cond  full_cond;
lock    full_cnt_lock;

void producer(void)
{
        int  new_msg;

while (true)  {
   new_msg=producer_new();
   lock(&empty_cnt_lock);
   while  (empty==0)
         cond_wait(&empty_cond, &empty_cnt_lock);
    empty-- ;
    unlock(&empty_cnt_lock);

      buffer_add_safe(new_mag);

      lock(&full_cnt_lock);
```

锁empty_cnt_lock：控制对变量empty的互斥访问/修改

条件变量：挂起wait()进程

基于条件变量和互斥锁的生产者-消费者问题

```
    lock(&full_cnt_lock);
    full++;
    cond_signal(&full_cond);
    unlock(&full_cnt_lock);
  }
}
```

锁full_cnt_lock：
控制对变量full的互斥访问/修改

条件变量：
唤醒进程signal()

基于条件变量和互斥锁的生产者-消费者问题

```
void consumer(void)
{
    int  cur_msg;
    while (true)  {
        lock(&full_cnt_lock);
        while  (full==0)
            cond_wait(&full_cond, &full_cnt_lock);
        full--;
        unlock(&full_cnt_lock);

        cur_msg=buffer_remove_safe();

        lock(&empty_cnt_lock);
        empty++;
        cond_signal(&empty_cond);
        unlock(&empty_cnt_lock);
        consume_msg(cur_msg);
    }
}
```

锁full_cnt_lock：
控制对变量full的互斥访问/修改

条件变量：
挂起wait()进程

锁empty_cnt_lock：
控制对变量empty的互斥访问/修改

条件变量：
唤醒进程signal()

Monitor vs Semaphore

- Monitors are implemented by semaphores
- For more details, *refer to* Appendix 6-2 Monitor Implementation Using Semaphores

- Monitors vs Semaphores
  - refer to next slide

信号量方式	管程方式
互斥访问的资源/变量, e.g.1 buffer e.g.2 shared data; readcount e.g.3 餐桌; state[i]	monitor; shared variables 说明：管程自动实现对管程所代表的资源或管程内shared variables的互斥访问，因此不需要单独定义类似于binary semaphores mutex角色的结构
binary semaphores: 对临界资源/变量互斥访问 e.g.1 mutex e.g.2 wrt, mutex e.g.3 mutex	
counting semaphores e.g.1 empty, full; e.g.2 self[i];	conditional variable x + 整型变量+互斥信号量
wait(), signal() on counting semaphores	x.wait(), x.signal() on conditional variables
concurrent processes的各个业务逻辑步骤	concurrent procedures
shared variables, binary semaphores, counting semaphores初始化	initialization code: 对shared variables，conditional variables、整型变量、互斥信号量的初始化 说明：无binary semaphores

Monitors vs Semaphores

- 以管程方式实现的并发进程，其业务逻辑由一系列顺序步骤组成；
- 当其中某些业务逻辑步骤需要访问临界资源时，需要采用定义在管程内的procedures；
- 如果进程在其生命周期内需要顺序访问多个临界资源，每个临界资源通过一个管程来控制对其互斥访问，则进程的业务逻辑步骤中会调用来自多个管程的procedures
  - e.g. “小和尚-老和尚挑水取水”

说　　明

- 软件同步适用于单CPU/单核系统
  - e.g. 反例：多/双核4线程系统上的Bakery同步

- 信号量/硬件/管程等其它几种机制适用于单、多CPU系统，特别是硬件指令、自旋锁
  - e.g. 多核系统中的随机数发生
  - e.g. “利用流水线和锁提高服务器吞吐量”

Appendix 6-2 Monitor Implementation Using Semaphores【略】

- Semaphores/Variables
  - semaphore mutex
    - initially = 1
    - a process must execute wait(mutex) before entering the monitor and must execute signal(mutex) after leaving the monitor
  - semaphore next, int next-count
    - initially, next = 0, next-count = 0
    - since a signaling process must wait until the resumed process either leaves or waits, an additional semaphore, next, is introduced, the signaling processes can use next to suspend themselves //用于控制与其它正在进入/离开管程的进程进行互斥
    - next_count is used to count the number of processes suspended

用
1) 二元信号量mutex、next
2) 整型变量next-count,
，控制管程内函数/过程的互斥执行

Appendix 6-2 Monitor Implementation Using Semaphores【略】

- Each external procedure *F* will be replaced by

```
wait(mutex);
    …
  body of F;
    …
if (next-count > 0)
      signal(next)
  else
     signal(mutex);
```

- Mutual exclusion within a monitor is ensured

Monitor Implementation Using Semaphores

- For each condition variable ***x***, we have:

  semaphore x-sem; // (initially = 0)

  int x-count = 0;

- The operation **x.wait** can be implemented as:

```
x-count++;
    if (next-count > 0)
        signal(next);
    else
        signal(mutex);
    wait(x-sem);
    x-count--;
```

用
1) 二元信号量x-sem; mutex, next
2) 整型变量x-count; next-count
, 实现管程内部的条件变量

Monitor Implementation Using Semaphores

- The operation **x.signal** can be implemented as:

```
if (x-count > 0) {
    next-count++;
    signal(x-sem);
    wait(next);
    next-count--;
}
```

- This implementation is applicable to the definitions of monitors given by both Hoare and Brinch-Hansen

Monitor Implementation Using Semaphores

- *Conditional-wait* construct: **x.wait(c);**
  - **c** – integer expression evaluated when the **wait** operation is executed
  - value of **c** (a *priority number*) stored with the name of the process that is suspended
  - when **x.signal** is executed, process with smallest associated priority number is resumed next

- Check two conditions to establish correctness of system:
  - user processes must always make their calls on the monitor in a correct sequence.
  - must ensure that an uncooperative process does not ignore the mutual-exclusion gateway provided by the monitor, and try to access the shared resource directly, without using the access protocols

Thanks for your attention